Stores Optimisation and Material Flow

Consolidated Summer Internship Project Report

COSMOS

Syamantak Industries Pvt. Ltd.

Inventory classification • Storage organisation • Physical stock rotation • Replenishment • Operational Kaizen

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Summer Internship Project

Submission date [To be completed]

Title Page

Stores Optimisation Material Flow and Operational Excellence

Table 1 Report particulars

Field	Detail
Organisation	COSMOS | Syamantak Industries Pvt. Ltd.
Facility name used in sources	Cosmos Air Distribution Products
Prepared by	Mahir Valani
Institution	Welingkar Institute of Management
Programme and specialisation	[To be completed]
Academic year and internship period	[To be completed]
Faculty guide	[To be completed]
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Purpose and scope

This report consolidates the Stores improvement programme into six main workstreams and six additional Kaizen initiatives. It explains the operating problem, intervention, evidence and controls for each, so management and academic reviewers can assess both the completed work and the proposed next steps.

Evidence convention

Implemented identifies a change reported as put into practice. Proposed identifies a design awaiting implementation. Measured refers to documented observations; reported results, estimated benefits and early observations retain their original qualifications. Recommended next steps are future actions.

Source S1, S2, S3, S4, S5, S6, S7, S8.

Chapter 1 Executive Summary

The Stores programme established a more structured physical material-management system at COSMOS while identifying the data and process controls needed to sustain it. Department-wise racks, suitable bins, location codes, labels, physical FIFO and a reference Kanban board address different parts of the same operating challenge: making the right material visible, accessible and available to Production.

The initial Gemba assessment recorded fragmented storage, materials on the floor, inconsistent identification, manual counting and stock-visibility difficulties. These were initial observations from one walk, rather than a verified plant-wide baseline. The process-mapping work separately documented manual GRNs and Excel-based issue and return tracking alongside ERP requisitions.

The strongest quantified evidence is the weight-based counting comparison: 14 min 58 sec before and 7 min 53 sec after for a batch of three MRS transactions, with approximately 47% reduction reported. The result is batch-specific and does not establish an individual-MRS saving or an error-rate improvement. The machine was already available, so no new equipment was required.

The Kanban report states that historical raw-material stockouts occurred approximately one to two times per month and that zero were reported after implementation. The subsequent observation period is unspecified and unaudited. The board is a reference implementation; extension to other Stores areas and digital Kanban remain future work. Supplier lead time is reported at approximately 20 days and is not shown to have shortened.

Other time benefits have weaker evidence. Binning and location organisation report approximate retrieval averages of 4–5 minutes before and 2–3 minutes after per material. Bin-box visualisation separately estimates 2–3 minutes less search time per MRS. These overlap operationally and must not be added. Kitting is proposed, with an expected Production-side benefit of approximately 2 minutes per job card; its net effect depends on the additional work in Stores.

VAV assembly was relocated from the second floor to the first floor with Production collaboration. Its source reports 18%, approximately 120 minutes per day, and other conflicting shift and percentage figures. The relocation is documented; a reliable quantified benefit requires reconciliation of the observation sheets and denominators.

Priority for management

Reconcile the FSN inputs and reorder formulas, record completed daily audits and Kanban events, validate comparable end-to-end timings, and obtain the mezzanine flooring specification and technical approval records. Maintain the physical controls while validating the proposed ERP GRN and MIN changes and a limited kitting pilot. No financial saving or combined efficiency percentage is supported.

Source S1, S2, S3, S4, S5, S7, S8.

Executive Scorecard

Table 2 Executive initiative scorecard

Initiative	Status	Evidence and reported impact
1 FSN	Analysis completed	Consumption-based prioritisation; data discrepancies remain.
2 Space and storage	Partial evidence	Existing rack and bin allocation documented; full rack-design dossier unavailable.
3 Binning and labels	Implemented	4–5 to 2–3 min per material; approximate reported averages.
4 Kitting	Proposed	About 2 min per job card expected in Production; net benefit unmeasured.
5 Physical FIFO	Implemented as reported	Defined old/new stock positions; no measured FIFO saving.
6 Kanban	Reference board implemented	Zero stockouts reported; subsequent period unspecified.
7 Weight counting	Implemented	14:58 to 7:53 for three MRS; approximately 47% reported.
8 SOPs and audit	Instructions issued; checklist developed	Revision 01 effective 01/07/2026; no completed audit results.
9 VAV relocation	Implemented	120 min/day and percentage claims conflict; verify.
10 Bin-box boards	Implemented	2–3 min/MRS estimated; no formal time study.
11 Kaizen recognition	Introduced	Two ideas received; early participation outcome.
12 Mezzanine floor	Implemented as reported	Wood replaced by steel; technical specification unverified.

How the interventions connect

Classification informs which items deserve accessible storage and replenishment attention. Storage and labels identify where material belongs. FIFO determines which stock to issue first. Kanban signals when to replenish. Counting, standard work, audit and worker participation sustain those controls. The VAV layout and mezzanine improvements address material flow and the working surface respectively.

Source S2, S4, S5, S6, S7, S8.

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Abbreviations

ERP enterprise resource planning; FIFO First In First Out; FSN Fast Slow Non-moving; GI galvanised iron; GRN Goods Received Note; MIN Material Issue Note; MRN Material Requisition Note; MRS Material Requisition Slip; P2P Procure-to-Pay; ROP Reorder Point; SIP Summer Internship Project; SKU stock keeping unit; SOP standard operating procedure; VAV variable air volume; VWI visual work instruction.

Chapter 2 Business Context and Improvement Framework

Stores supports manufacturing by receiving, locating, counting, rotating and issuing materials, and by communicating replenishment needs to Purchase. The programme covers Factory 1 and Factory 2 Stores, extrusion, spares and powder materials, and the VAV flow described in the layout work. It does not establish a quantified baseline for the whole plant.

Diagnostic sequence and operating context

The Gemba header records 07/05/2026 for Factory 1 and 2; its document date is 15/05/2026. The process-map report is dated 20/05/2026 and the FSN report 23/05/2026. These dates describe the individual source documents or activity, and do not establish the full internship duration. The diagnostic visited shop-floor material use before Stores, captured issues and prioritised them by severity. The exact feedback mechanism and severity criteria remain undocumented.

Initial opportunity areas

The walk identified same materials held in different locations, unstructured racks, floor storage, long-held stock, inconsistent labels, stock visibility and stockout difficulties, and manual counting. Point-of-use storage, structured 5S, FIFO, Kanban, binning and kitting were not evident during the walk. This means they were not seen, rather than proving that no such practice existed elsewhere.

Table 3 Programme relationship diagram

Connected intervention	Contribution to the overall objective
FSN → storage prioritisation	Accessible locations for high-consumption items; review of non-moving material.
Rack allocation → binning → labels	Defined department, rack, level and position; material and quantity visibility.
Physical FIFO → material issue	Old stock first through physical separation and identification.
Kanban → Stores → Purchase → receipt	Visible reorder status and replenishment follow-up.
Kitting proposal → Production readiness	Consolidate SKU materials before handover; measure the added Stores effort.
Counting + SOPs + audit + participation	Repeatable operation, discrepancy reporting and corrective action.
VAV layout + mezzanine flooring	More direct inter-floor flow and an improved working surface.

Figure 1 Overview of the connected improvement framework

Editorial overview of documented relationships; arrows do not imply measured causality.

Source S1, S2, S3, S4.

Material Management Process and System Boundary

The current process combines ERP, manual receipt documentation and an Excel file on one PC used by two storekeepers. MRN and MRS serve different purposes: Stores raises the Material Requisition Note for procurement; Production raises the Material Requisition Slip for issue against a job.

Table 4 Material management current and proposed flows

Current process	Improved or proposed process
Customer order → Sales Job Card in ERP → Stores determines requirement and raises MRN → P2P procurement → receipt and manual GRN → storage. Separately: Production MRS in ERP → issue → Excel quantities, extra issues, rates and returns.	Proposed receipt: physical verification → GRN in ERP → inward stock update. Proposed issue: MRS → availability check → material issue → MIN in ERP → outward stock update. Stock updates and reporting require ERP validation.

Returns and digital transition

The current 15-day policy applies to unused material returned from Production, with manual Excel adjustment. Approval, documentation and exceptions require confirmation. No ERP returns workflow is assumed. Opening-stock reconciliation, transaction capability, role definition, training and the treatment of extra issues and rates must be resolved before transition.

Physical and digital controls

ERP GRN and MIN creation are proposals, and the ROP check, alerts and stock visibility depend on available configuration. A digital stock record does not itself implement physical FIFO. Similarly, handwritten quantity fields and printed rack boards do not update ERP automatically. Original process maps are retained in the visual evidence appendix; written process clarifications govern any conflicting visual detail.

Source S3, S8.

Chapter 3 Main Stores Optimisation Workstreams

1 FSN Inventory Classification

The objective was to prioritise material accessibility and inventory control using six months of consumption data. The FSN report records cleaning historical Stores data, ranking items by consumption and grouping them by cumulative share. It preserves Fast (A), Medium (B), Slow (C) and Non-moving (NM), including the additional Medium band actually used. The legacy A/B/C codes do not convert this into value-based ABC analysis; no XYZ analysis is demonstrated.

Table 5 FSN method and classification thresholds

Store	Documented class boundaries	Traceability
Extrusion	Fast reaches 68.3% (threshold not stated); Medium to 90%; Slow to 100%; no NM shown.	Typed dashboard; no family-level register.
Factory 1 Spares	Fast 0–70%; Medium 70–90%; Slow 90–100%; NM zero consumption.	51 representative SKUs supplied within the source account of 549.
Powder	Fast top 80%; Medium next 15% to 95%; Slow remainder; NM zero consumption.	Source report recomputes 56 shades; cumulative threshold retained.

The documented Powder reproduction places an item in a class only while cumulative share stays within the threshold: three Fast shades reach 75.7%, and eleven Fast plus Medium shades reach 94.6%. Equal-consumption shades at the boundary have no stated tie-break rule. This is the method reported in the source, rather than a transaction-frequency or days-since-last-issue formula.

Table 6 FSN category counts and concentration

Store and scope	Fast	Medium	Slow	Non-moving
Extrusion 118 families	12; 68.3% consumption	19; 21.7%	87; 10.1%	Not shown
Spares 549 stated SKUs	19; ≈69.3% active consumption	31; ≈20.6%	329; ≈10.1%	169; 30.8% of SKUs
Powder 56 shades	3; 75.7% recomputed	8; 18.9%	20; 5.4%	25; 44.6% of shades

Spares counts sum to 548, while the stated total is 549. Powder workbook shares are 76.2%, 18.9% and 4.9%; the source report recomputes 75.7%, 18.9% and 5.4%. Extrusion rounded shares sum to 100.1%. Figures remain as reported.

Source S2.

FSN Stock Parameters Findings and Controls

Extrusion average monthly consumption is reported at 92,352 pcs: Fast 63,035, Medium 20,028 and Slow 9,289. For Spares, the 51-SKU subset totals 4,651,455 over six months; its 19 Fast items total 3,898,794. Grille Black Bush and Pop Rivet 1/8 × 1/4 average 149,712 and 100,669 respectively per month. The FSN source flags inconsistent unit labelling for Spares; these quantities must not be compared with Powder weight or assigned a monetary value.

Powder consumption totals 19,384 in the source report's raw-data recomputation, versus 18,844 in the underlying summary. RAL 9016 Matt and Satin Black account for 70.2% together; 13 of the 20 Slow shades moved in only one of six months. Zero movement is a review trigger and does not by itself prove that a spare is obsolete or non-critical.

Table 7 Stock control rules as documented

Store	Average and buffer calculation	Reorder trigger
Spares	Monthly = ROUND(AVERAGE(six months),0); daily = ROUND(monthly/30,0); safety = daily × 10.	ROP = daily × 30, interpreted as 20 lead-time days + 10 safety days.
Powder	Monthly and daily rounding as above; safety = daily × 3.	ROP = daily × 13, interpreted as 10 lead-time days + 3 safety days.
Extrusion Medium	Safety stock = 1.5 × average monthly consumption, stated as text.	No formula-based ROP or Max value supplied.

For Grille Black Bush, daily consumption 4,990 gives safety stock 49,900 and register ROP 149,700; a conflicting source summary gives ROP 74,850 using 15 days. For RAL 9016 Matt, daily 41 gives safety stock 123 and ROP 533. Day multipliers are hard-coded, so changing lead-time input columns does not update the formula. Rounding low Powder averages can give zero daily consumption and zero ROP for shades with average monthly consumption of 14 or less.

Operational implications and next steps

Place Fast items in accessible, controlled zones; retain ROP monitoring for Fast and Medium; review Slow items for order-on-demand and Non-moving items for technical disposition. Review the 169 Spares and 25 Powder shades before any return, substitution or write-off. Confirm machine dependency independently of movement class. Validate supplier lead times, buffers, thresholds, ties and rounding before using these parameters for Kanban sizing. The source's 30%+ rack-space, 15–20% SKU and 30–40% shade-reduction ideas are unvalidated hypotheses, not outcomes.

Source S2.

2 Store Space and Storage Design

The diagnostic identified floor storage, fragmented material locations and inadequate structured storage. The available binning report documents a practical redesign using existing racks: usable length, depth and height were assessed, bin sizes selected for fit and access, and racks allocated by department. This establishes storage organisation but does not substantiate a complete new-rack capacity study.

Design implemented in the available evidence

One department was assigned to each rack so an operator could go directly to the relevant material family. Lighter materials were placed on upper levels and heavier materials at the lowest levels, subject to safe handling. Rack numbers, levels and positions support traceability. Spare-parts FIFO uses two stock positions within existing bins where feasible because extra bins and capital were constrained. Bulky extrusion uses separate physical identification suited to its form.

Table 8 Recorded bin dimensions and order quantities

Prefix	Length	Breadth	Height	Bins ordered
—	275	400	400	72
—	295	400	400	24
—	345	400	400	16
—	540	360	400	12
CC	300	200	200	36
CC	400	300	320	30
CH	540	360	400	12
CH	650	450	315	15

Total bins ordered: 217. Dimension units and the CC/CH prefix meanings require confirmation. An order quantity is not a verified installed storage capacity.

Coverage and verification boundary

No separate space-and-rack-design report is among the eight supplied documents. The FACT framework, factory- and floor-wise storage demand, rack dimensions, load ratings, levels × positions capacity calculations, capacity gaps, non-rack floor allocations and expansion layouts are not documented sufficiently to reproduce. They remain requiring verification; no capacity or recovered-space claim is made. The available factory floor drawings relate to the VAV relocation and are retained under that initiative.

Retain the department allocation, weight-based vertical placement and location identity. Validate actual fit, access and bin dimension units against the recorded order sheets before extending the arrangement. Do not use the number of bins ordered as proof that all were installed or that a floor-wise capacity deficit was closed.

Source S1, S4, S6, S8.

3 Binning Labelling and Storage Location Optimisation

The implemented arrangement connects physical storage to a clear identity: rack measurement → suitable bin selection → department allocation → weight-based placement → location coding → standard labels → manually updated quantity display. The purpose is to reduce searching and unnecessary walking, reaching and handling.

Table 9 Location and material identity

Field	Documented example or use
Physical location	RD-R5-L2-P2: department prefix RD, Rack 5, Level 2, Position 2.
Software code	SI/SP-201.2 identifies the Round Diffuser Rod item in the system. It is separate from the location code.
Item name	Round Diffuser Rod – NS 225 – 97 mm.
Minimum quantity	500 on the photographed rod label; 6000 on the Linear Grille Frame example.
Current quantity	791 is visible on the rod photograph; it is a historical displayed value, not a current inventory claim.

Rack No. 5 carries level boards listing item names and software codes. The Level 2 board places the rod in its second entry, consistent with Position 2. Item labels use five fields: item name, software code, minimum quantity, current location and current quantity. The Linear Grille Frame example uses SI-EX-10.1; blank photographed fields are not inferred.

Label generation and quantity control

A Word A4 label template was connected to structured Excel material data, previewed, printed and attached after identity and location verification. Mail Merge is the probable method reported; the exact commands are not recalled. A white magnetic writable surface on GI sheet allows quantity to be erased and rewritten without reprinting fixed identity fields. Updates are manual and require reconciliation with physical stock and inventory records.

Evidence of impact and control

Retrieval is reported at approximately 4–5 minutes before and 2–3 minutes after per material, from MRS receipt through displayed-quantity update. These are approximate operating averages, not stopwatch records. The source derives about 2 minutes and roughly 40–45% from range midpoints; this remains indicative. Validate a sample of comparable MRS retrievals and observe walking, searching and lifting. Maintain visible labels and accurate quantity and location fields through the daily audit.

Source S8.

4 Kitting Improvement

The existing MRS already specifies the raw materials needed for the finished-product SKU. Stores picks them from their locations but issues them separately. Production then identifies, differentiates and organises the received items. The opportunity is to consolidate the specified materials before handover.

Table 10 Kitting current and proposed processes

Current process	Improved or proposed process
Production submits MRS → Stores reviews and picks required materials → items issued separately → Production identifies and organises materials → manufacture.	Proposed: MRS → Stores picks → consolidates a SKU kit → verifies contents against MRS → identifies kit by SKU and MRS → hands over → Production uses grouped materials.

Proposed process design

Use a suitable bin or container where dimensions, compatibility, weight and handling requirements permit. If a single container is unsuitable, a clearly identified group of containers is a design option. Kit verification and identification are proposed controls, rather than confirmed existing practice. The original source diagrams are retained in the visual evidence appendix.

Expected impact and net workload

The reported expected saving is approximately 2 minutes per job card in Production, arising from less identification and organisation after receipt. There is no recorded before-and-after study. Kitting adds consolidation and verification work in Stores, so the total process saving may be smaller. The estimate belongs to kitting and is not a FIFO benefit.

Table 11 Kitting pilot evaluation

Measurement	Purpose
Before: Production identification and organisation per job card	Establish the claimed Production-side baseline.
After: Stores picking plus kit consolidation and verification	Capture additional Stores workload.
Before and after: total material-readiness time for the same SKUs	Test the net end-to-end change.

Net time effect = Production-side time saved − additional Stores kitting time, measured per job card on comparable SKUs. Pilot selected SKUs, maintain SKU/MRS identification and prevent cross-job mixing before wider rollout. Implementation is not confirmed.

Source S7.

5 Physical FIFO Stock Rotation

Before the intervention, stock age was identified largely by dust: dusty material was treated as old, and clean material as new. That judgement depends on appearance rather than a defined separation. The FIFO work introduced physical old/new positions using existing bins and available waste GI sheet, within space and capital constraints.

Table 12 Physical FIFO current and improved processes

Current process	Improved or proposed process
Approach stock → inspect appearance → infer old/new from dust → select and retrieve for issue.	Identify front or red old stock → issue old first → when exhausted, lift divider where applicable and bring rear stock forward → reset positions → place the next receipt in the rear or green new-stock position.

Table 13 FIFO configuration by material

Material category	Physical arrangement documented
Spares in both factories	GI-sheet divider within the bin: front old stock, rear new stock. Divider handled when the old section is empty, not at every issue.
Bulky extrusion	Red old-stock and green new-stock identifiers using GI-sheet pieces, with Old/New labels.
Actuators	Red/green GI-sheet identifiers; left-to-right arrangement reported.
Spigote and DCV; coils; blowers	Red/green GI-sheet identification reported.
ArmaSound and ArmaFlex	Red/green spray identification reported.

Implementation and evidence limits

The arrangements are reported as introduced and photographs support the spare-bin configuration and staff briefing. The source still flags category-level configuration verification. Its handling observations—Rod 1.5–2 min, Corner 2 min, Screw 2 min, R/D rod 1 min, Rivets 4–5 min and Side spacer 5 min—do not identify a comparable before/after FIFO activity. They are not savings and support no FIFO improvement percentage.

VWI-01, Revision 01 effective 01/07/2026, directs the Foreman to maintain the sequence and update red/green magnets, with Assistant Manager daily inspection. Confirm how these magnets correspond to GI dividers, tags and spray by material. Keep labels visible, segregate damaged or unidentified stock, and audit receipt placement, first-issue selection and resets. No measured time, space, cost or safety gain is claimed.

Source S4, S6.

6 Kanban Based Inventory Replenishment

The source reports approximately one to two raw-material stockouts per month and an internal audit finding physical stock below recorded stock. Kanban addressed low-stock visibility and replenishment communication; physical reconciliation and correct quantity updates remain separate inventory controls.

Interim communication and reference board

The Stores Assistant Manager first compiled quantities and reorder requirements and emailed the Stores Manager, who coordinated with Purchase and followed receipt. This was a manual interim process. The physical reference board subsequently introduced rows for GRILLE, SLOT & HCS, DIFFUSER and OBD, with a status legend and Daily Kanban Review fields for date, reviewer, pending cards and critical cards.

Table 14 Kanban statuses and required action

Status	Documented meaning
Green available	Material available; no immediate action. Confirm whether the card stays at the material location or in the green column.
Yellow reorder	Reorder point reached; initiate replenishment communication.
Red critical or urgent	Immediate action and escalation. Red is a priority designation, not a mandatory stage for every item.
Blue in process	Request or order actually raised; follow up until receipt. Return to green when material is available.

Workflow and responsibilities

Operator updates quantity and compares it with the applicable minimum/ROP → yellow when reorder is needed, red if critical → Assistant Manager compiles requirements → Stores Manager coordinates Purchase → Purchase processes the order → blue reflects actual progress → receipt and quantity update return the status to green. Card movement, communication, purchase processing and receipt are distinct events.

Each card records material name, software code, rack location, department and supplier, often with an extrusion profile. The source describes 112 External Supply cards; three examples are embedded and the full register is separate. No minimum or ROP field is on the card itself. Board rows must be reconciled with fifteen card department labels; six duplicated software codes and the section-not-found note for SI-EX-37.3 require review.

Zero stockouts are reported after implementation over an unspecified, unaudited period. Card movement and communication take approximately 2–5 minutes by estimate; procurement lead time is approximately 20 days. Neither is a measured time reduction. Extension beyond the reference board and digital Kanban are planned. Define thresholds, reconcile quantities and record reorder events before transition. VWI-02 omits the board's blue stage; align the procedure and board before rollout.

Source S4, S5.

Chapter 4 Additional Kaizen and Operational Improvements

7 Weight Based Counting

Small spares—screws, bushes, spacers, aluminium rods and grille hardware—were counted piece by piece during MRS preparation. Counting effort therefore increased with the quantity. The implemented method uses the existing Gill Real Weigh digital weighing machine in piece-count mode, with stated 4 kg capacity. A precisely hand-counted sample of 10 pieces establishes reference unit weight, after which the batch weight yields a displayed count.

Table 15 Counting current and improved processes

Current process	Improved or proposed process
MRS requires spares → pick batch → count each piece manually → issue.	Setup per spare type: hand-count 10 → weigh sample → establish unit weight. Batch: pick → place on scale in count mode → read count → issue.

Table 16 Recorded batch counting comparison

Measure	Before	After or outcome
Time for three MRS transactions	14 min 58 sec	7 min 53 sec
Saving for that batch	—	7 min 05 sec
Cycle-time reduction	—	Approximately 47% as reported
Additional capital equipment	Existing scale available	Nil additional equipment reported

Table 17 MRS transactions identified in the counting source

MRS reference	PPBOM	Date
SI/MRS/2026-27/1196	PPBOM_15292	28/05/2026
SI/MRS/2026-27/1239	PPBOM_15335	30/05/2026
SI/MRS/2026-27/1265	PPBOM_15361	01/06/2026

The method is reported as standardised across Factory 1 and Factory 2 Stores for applicable bulk hardware. Stopwatch photographs support one three-transaction batch comparison, not a per-MRS result or a general productivity rate. Confirm whether identical transactions and quantities were used for both readings. Count accuracy is not quantified; the certification description in the source is not a calibration record.

Record reference weights, check samples periodically and repeat timings across more batches. Part-weight variation affects reliability; the 4 kg limit constrains a single placement. Record discrepancies to test accuracy and avoid extrapolating this one batch to all MRS types.

Source S4.

8 Bilingual SOPs and Daily Store Audit

Standardised visual instructions were issued in English and Marathi to support consistent FIFO and replenishment routines. VWI-01 FIFO and VWI-02 Kanban are Revision 01, effective 01/07/2026. The Daily Store Audit Checklist was developed with fifteen checkpoints, yes/no responses, remarks, audit summary, corrective actions, comments and sign-off. Duration and extent of shop-floor use are not documented.

Table 18 Daily Store Audit Checklist checkpoints

No	Checkpoint	No	Checkpoint
1	FIFO followed during material issue	9	Machine counting completed for newly received stock
2	New stock stored behind existing stock	10	New stock quantity verified before storage
3	Red and Green magnetic labels correctly updated	11	Kanban board updated wherever minimum stock reached
4	Current Quantity updated on magnetic sheet	12	No unidentified or loose material found
5	Current Location updated correctly	13	Store area clean and organised 5S
6	Material stored in designated rack/location	14	Damaged materials identified and segregated
7	Rack labels visible and readable	15	Stock discrepancy reported to Assistant Manager
8	Item labels properly attached	—	—

Responsibility and corrective action

The Foreman or Store Keeper checks stock daily, updates displayed quantities and indicators, maintains FIFO and informs Purchase at yellow. The Assistant Manager checks compliance and follows replenishment and corrective actions. The checklist records Total Checkpoints 15, Complied, Non-Complied and Compliance %, with Issue Identified, Action Taken, Responsible Person and Target Date. Foreman/Auditor and Assistant Manager sign and date the record.

The original checklist and all four language sheets are reproduced in the appendix. The Marathi sheets follow the English layout and principle but are not word-for-word translations. They are retained as original visuals to preserve the wording and visual instruction. No completed checklists support a measured compliance, error or productivity change.

Retain completed records, track non-compliance and action closure, and review revisions. Reconcile VWI-01 magnets with the category-specific FIFO arrangements and VWI-02's three-colour instruction with the board's four statuses. Keep minimum/ROP wording and item-code examples aligned with actual material records.

Source S4, S5, S6.

9 VAV Assembly Line Relocation

The earlier observed route moved from the ground floor to second-floor VAV assembly, returned to the first floor and then to ground-floor dispatch. The VAV line was relocated from the second floor to the first floor through an AutoCAD layout prepared in collaboration with Production. The objective was to remove unnecessary inter-floor movement within the available area.

Table 19 VAV material movement comparison

Current process	Improved or proposed process
Ground floor → second-floor VAV assembly → first floor → ground-floor dispatch. Activities along the original route require drawing validation.	Ground floor → first-floor VAV assembly and related operations → ground-floor dispatch. The expected route requires confirmation against final drawings.

The first-floor drawing places VAV body and flange clinching, wiring and control-box assembly, calibration, store and final assembly check/packing on the first floor, with hand-drawn arrows. The second-floor drawing shows damper and diffuser assembly. Original, first-floor, ground-floor and second-floor drawings are retained without reconstructing their geometry. The original photographed sheet has limited label legibility.

Table 20 VAV reported figures requiring reconciliation

Source location or basis	Figure retained	Verification issue
Narrative and chapter overview	18% reduction; approximately 120 min/day out of 480 min or 8 hours	18% is not explained by 120/480.
Time and motion table	540 min or 9 hours standard shift	Conflicts with the narrative 480-minute basis.
Hourly saving in table	15 min/hour calculated as 120 ÷ 8	Uses an eight-hour basis.
Saved share of day in table	25%, with calculation shown as 120 ÷ 475	Percentage and denominator do not agree precisely.
Reduction and closing status	25.26%; closing text uses 25,26%	Conflicts with 18%; no observation sheets supplied.

The relocation is implemented as reported; the percentages and daily saving are not independently validated and are not interchangeable. Obtain the observation sheets, define the measured material-handling activity and operator count, reconcile the working-day denominator, and record travel distance and handling time for comparable VAV orders. Retain the agreed layout and mark validated routes on clear drawings.

Source S4.

10 Bin Box Visualisation

Rack-level boards were implemented so operators could identify material without searching multiple bins and locations. The improvement uses the existing racks and blue bins. A rack sign identifies the rack and department—for example Rack No. 2 Grille—and each level board lists item names and software codes.

Table 21 Bin box identification process

Current process	Improved or proposed process
MRS → locate rack and level information → search multiple locations where information is unclear → retrieve material.	MRS → identify rack sign → read level board item name and software code → match the bin label → retrieve material.

The shared software code connects the board reference with the physical bin. The physical location code remains a separate address. The boards are printed references and are not stated to update automatically from ERP. The original walkthrough stills show level boards and labelled bins; an earlier-condition photograph shows racks without the consolidated level boards.

The reported reduction is an estimated 2–3 minutes of search time per MRS, with no formal before-and-after study. This differs from the binning report's approximate time per material and must not be combined with it. Material-search steps overlap, and daily MRS volumes and the calculation basis are not available.

Define an owner for board updates, maintain matching codes when items move, and measure a sample of comparable MRS retrievals with and without boards. Keep board content readable at the point of use and include location and label checks in the daily audit.



Figure 2 Rack and level material boards in use

Source S4. Original visual retained; source resolution governs legibility.

Source S4, S8.

11 Worker Participation Through Kaizen Recognition

Prizes and Worker of the Month recognition were announced to encourage workers to contribute improvement ideas. The earlier participation process and baseline submission rate are not documented, so the result cannot establish a sustained engagement increase.

Table 22 Two Kaizen ideas reported as received

Idea	Description and scope
Month-end monetary reward for Kaizen	Money and a bonus for an operator contributing Kaizen, paid at month end as described. Award value is not documented.
Operator of the Month	Recognition in each department, with Production, Stores, Factory overall and Dispatch named in the source.

The source uses Worker of the Month for the announced recognition and Operator of the Month for the second received idea. Those terms are preserved rather than treated as evidence of separate completed award programmes. Implementation of each idea and payment of rewards are not confirmed.

The two received ideas are an early participation outcome. No worker names, award amounts, savings, productivity gain or participation rate is supported. The source recommends a simple idea record: title, area, current situation, proposed improvement, expected benefit, submitter/date, review decision, recognition and implementation status. Track ideas submitted, reviewed and implemented, time to review and verified benefits. Keep the review and recognition transparent through those records.

Source S4.

12 Mezzanine Floor Safety Improvement

The wooden mezzanine flooring was replaced with steel flooring for worker safety, as reported. The supplied before photograph shows a local broken and splintered surface with a visible gap. The walkthrough and current-floor photograph show the mezzanine and red-tape marking. They do not establish structural suitability or a quantified safety improvement.

The technical specification, load rating, slip resistance, installation method, inspection requirements and engineering or safety approval are not established. No incident statistics, accident reduction or certification are claimed. Record these technical details, obtain appropriate inspection and approval records, add a clear steel-floor close-up, and include the surface in periodic inspection. Before and current-condition photographs are retained in the appendix.

Source S4.

Chapter 5 Consolidated Results and Evidence

Table 23 Consolidated results main workstreams

Initiative	Main intervention	Status	Reported result	Evidence type	Limitation and follow-up
FSN	Consumption ranking and review lists	Analysis completed	12/118 Extrusion, 19/549 Spares and 3/56 Powder Fast items	Analytical findings	Reconcile counts, totals, formulas and full registers.
Space and storage	Existing-rack fit and department allocation	Partial documented design	217 bins ordered	Recorded order quantities; qualitative design	Units, installed capacity and full rack-design study unverified.
Binning and labels	Location coding and writable quantity labels	Implemented	4–5 to 2–3 min per material	Approximate reported averages	Comparable stopwatch study and record reconciliation.
Kitting	SKU grouping and MRS verification	Proposed	≈2 min/job card in Production	Estimated benefit	Measure added Stores effort and net end-to-end time.
Physical FIFO	Front/rear divider; red/green identification	Introduced as reported	Defined issue-first method	Qualitative and photographs	No measured FIFO saving; verify category configuration.
Kanban	Board, cards and replenishment communication	Reference implementation	1–2 stockouts/month before; zero subsequently	Reported early observation	Post-period unspecified; retain event and replenishment logs.

These measures cover different activities and evidence types. They cannot be added into an overall efficiency percentage, annual time saving or financial benefit. Inventory accuracy requires reconciliation evidence independently of replenishment visibility.

Source S2, S5, S6, S7, S8.

Consolidated Results Additional Initiatives

Table 24 Consolidated results additional initiatives

Initiative	Main intervention	Status	Reported result	Evidence type	Limitation and follow-up
Weight counting	Reference sample and existing scale	Implemented	14:58 to 7:53 for three MRS; ≈47% reported	Stopwatch batch observations	Repeat comparable batches; verify same transaction basis.
SOPs and audit	Bilingual VWIs and 15-point checklist	Issued/developed	Common written and visual standard	Original control documents	No completed audit series; record compliance and closure.
VAV relocation	Assembly second floor to first floor	Implemented	120 min/day; 18% and conflicting figures	Reported result; drawings	Reconcile shift, denominator, operators and observation sheets.
Bin-box boards	Rack-level item/code reference	Implemented	2–3 min/MRS less search time	Reported estimate; stills	Formal time study absent; avoid overlap with binning.
Kaizen recognition	Recognition and idea submission	Introduced	Two ideas received	Early reported outcome	Reward values and implementation status unconfirmed.
Mezzanine floor	Wood replaced with steel	Implemented as reported	Improved working surface intended	Photographs; reported replacement	Specification, load rating and technical approval needed.

Validation priorities

Retain the original stopwatch/MRS evidence for counting; define dated stockout and reorder logs for Kanban; time comparable retrieval and kitting cycles; reconcile VAV observation sheets; retain completed audit checklists and closed actions; and obtain the mezzanine inspection and flooring records. These are the verification actions already identified across the source reports.

Source S4.

Chapter 6 Implementation Governance and Sustainability

Sustaining the programme depends on consistent physical operation and credible records. The following controls consolidate source responsibilities and recommendations. They are not evidence that every routine is operating or that named role-holders have approved a new governance structure.

Table 25 Governance and operational control

Control	Documented role or responsibility	Record and review
FIFO sequence and receipt placement	Foreman maintains positions and red/green indicators; Assistant Manager verifies.	Daily inspection and checklist; damaged/unidentified stock escalated.
Kanban quantities and status	Foreman/Store Keeper updates quantity and moves status; Assistant Manager checks; Stores Manager coordinates Purchase.	Daily Kanban Review; pending/critical cards; replenishment follow-up.
Locations and labels	Stores operator updates writable fields; board-update owner to be defined.	Audit location accuracy, labels and designated racks; reconcile physical stock and records.
Machine counting	Stores operator uses reference sample; Assistant Manager receives discrepancies.	Sample-weight record; periodic checks; new-receipt count/quantity verification.
Audit and corrective action	Foreman/Auditor records findings; Assistant Manager reviews and follows closure.	15 checks; remarks; action, responsible person and target date; comments and sign-offs.
FSN review and stock parameters	Workbook owners resolve data; Purchase confirms lead-time/buffer assumptions.	Medium Extrusion bi-weekly review; Slow/NM quarterly review as recommended; formula audit.
Training and adoption	Stores, storekeepers and Production roles to be defined for ERP and pilots.	Use bilingual instructions; revision review; clarify handover and transaction responsibilities.
Kaizen participation	Idea-review and recognition responsibilities to be agreed.	Submission, review, recognition and implementation record; verified benefits.
Flooring and revised VAV flow	Technical approval/inspection role to be confirmed; Production collaboration documented for layout.	Floor specification and periodic checks; validated layout routes and handling observations.

Recommended implementation sequence

Maintain the physical system and retain daily control records. Resolve FSN and stock-parameter discrepancies with the source owners before expanding Kanban. Validate ERP GRN/MIN functionality, opening stock and the returns procedure. Pilot kitting on selected SKUs and record net time across Stores and Production. Extend the board or digitise its statuses only after thresholds, department mapping and operating routines are confirmed.

Source S2, S3, S4, S5, S6, S7, S8.

Verification Register and Source Traceability

Table 26 Material issues requiring verification

Area	Issue to close
Scope and academic details	Complete programme, dates and guide fields. No separate rack-design dossier or FACT details available.
FSN coverage	Cover says four workbooks including Others; analysis documents three. Obtain Extrusion item register and full 549-SKU Spares register.
FSN numerical consistency	Resolve 548/549 Spares count, 15/30-day Spares ROP, Powder 825/1,365 month 2 and 18,844/19,384 totals; preserve stated/recomputed shares.
FSN method and parameters	Confirm threshold differences, equal-quantity tie, unit labels, rounded daily quantities, hard-coded formulas, Powder grand-total average 98, shade name Satin Black 9005/944 and independent spare criticality.
Physical and digital records	Verify bin units, installed capacity, board/bin/location correspondence; ERP stock updates, returns and opening balances.
FIFO and Kanban	Confirm category configurations, magnet/tag/divider correspondence, green-card location, blue stage, minimum/ROP wording, card duplicates and department mapping.
Outcome validation	Counting comparability; dated Kanban stockout window; formal retrieval study; VAV denominator and observation sheets; net kitting time.
Sustainability and safety	Retain completed audits, confirm reward/idea status, flooring specification, load suitability and inspection/approval.

Source register

S1 Gemba Walk & Initial Diagnostic Assessment — Store Optimization Project (1).docx

S2 Step 3 — FSN Inventory Analysis & Inventory Rationalization.docx

S3 Step 2 — AS-IS & TO-BE Process Mapping — Factory Material Management.docx

S4 Additional_Kaizen_Initiatives_SIP_Chapter_COSMOS.docx

S5 Kanban_Replenishment_SIP_Report_COSMOS.docx

S6 FIFO_Improvement_SIP_Report_COSMOS.docx

S7 Kitting_Improvement_COSMOS.docx

S8 Binning_Labeling_Storage_Location_Optimization_COSMOS.docx

Source references identify the supplied DOCX reports. Workbook recomputations and audit findings are those reported within S2; the underlying Excel files were not supplied for this consolidation. S1, S2 and S3 identify themselves as first drafts, and S1 observations remain pending validation. These qualifications are retained when using their evidence.

Appendix A Original Visual Evidence

The following exhibits preserve operational details that are best read in the source photographs, layouts, charts and work instructions. They retain original proportions and content. Repeated decorative images are omitted. Low-resolution source labels are not reconstructed. References in each caption identify the source report.

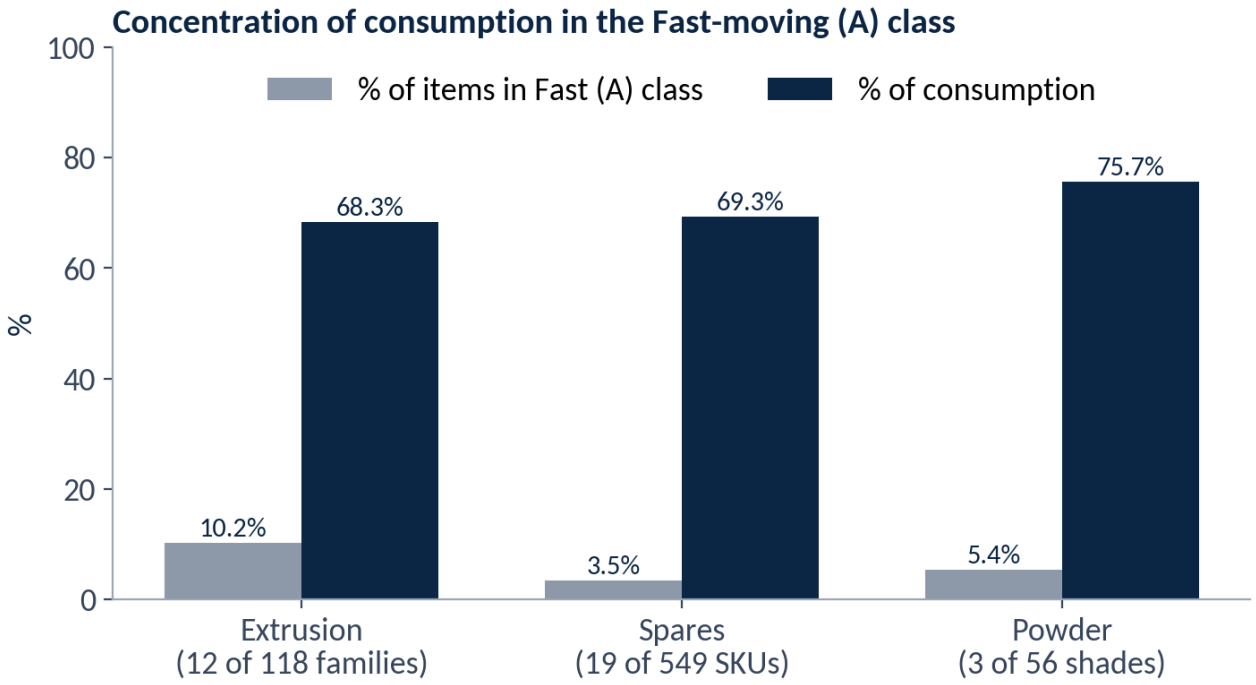


Figure 3 Fast class consumption concentration across the three stores

Source S2. Original visual retained; source resolution governs legibility.

The chart is an analytical result from S2, including its recomputed Powder share. Spares counts and shares remain subject to the discrepancies identified in Chapter 3. A shared chart does not imply that pieces, spare quantities and Powder weight can be totalled.

Diagnostic Photographs



Figure 4 Observed storage condition photograph one

Source S1. Original visual retained; source resolution governs legibility.



Figure 5 Observed storage condition photograph two

Source S1. Original visual retained; source resolution governs legibility.



Figure 6 Observed storage condition photograph three

Source S1. Original visual retained; source resolution governs legibility.

Material Management Current and Proposed Maps

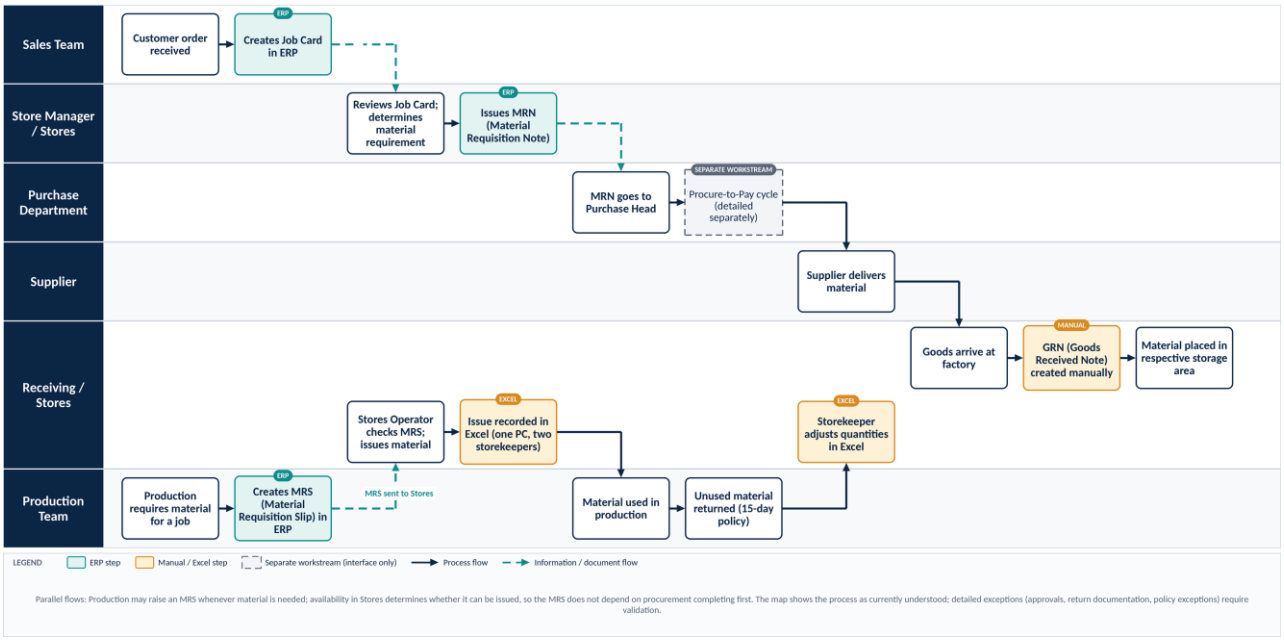


Figure 7 Current material management swimlane map

Source S3. Original visual retained; source resolution governs legibility.

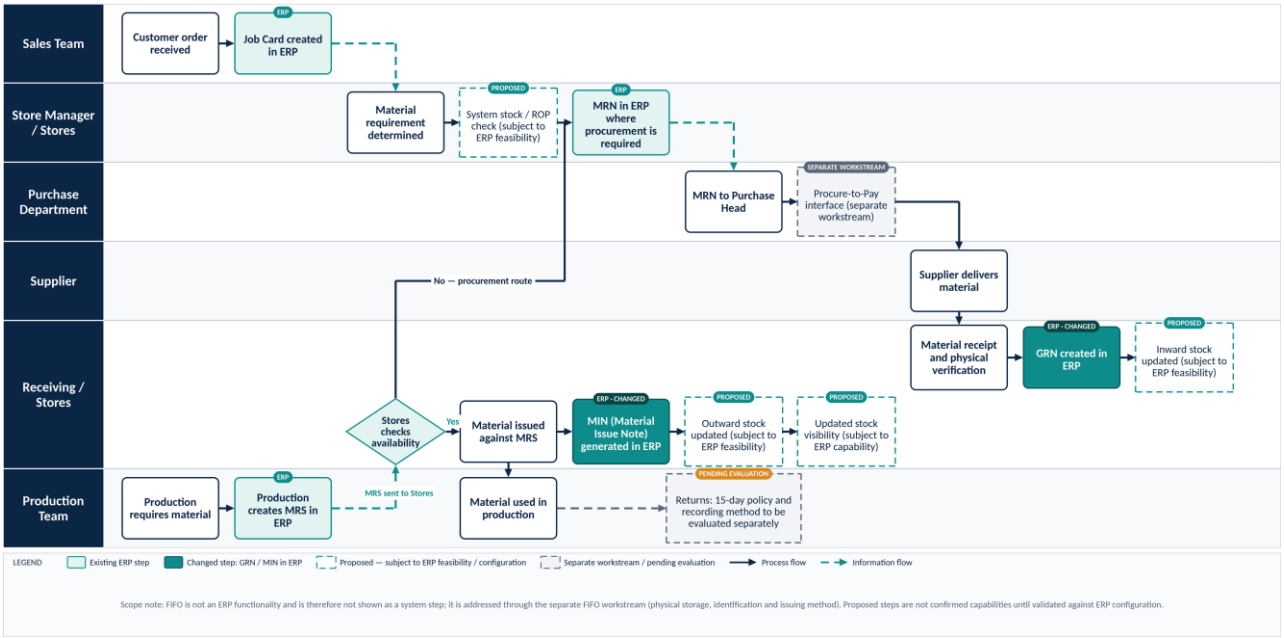


Figure 8 Proposed material management swimlane map

Source S3. Original visual retained; source resolution governs legibility.

Material Management Supporting Reference Maps

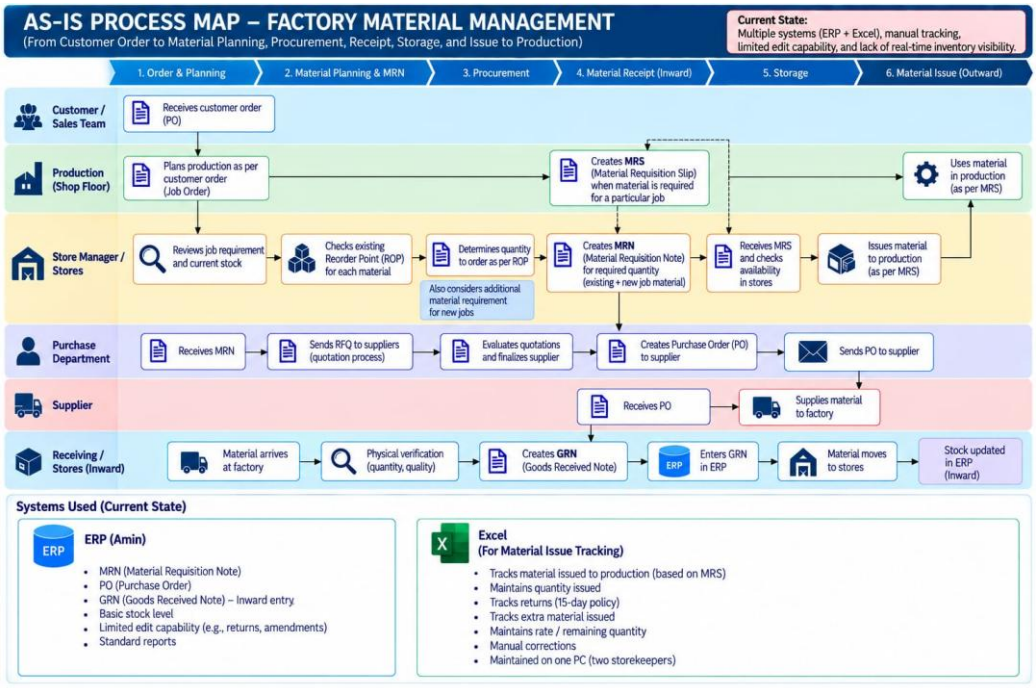


Figure 9 Source reference current material management map

Source S3. Original visual retained; source resolution governs legibility.

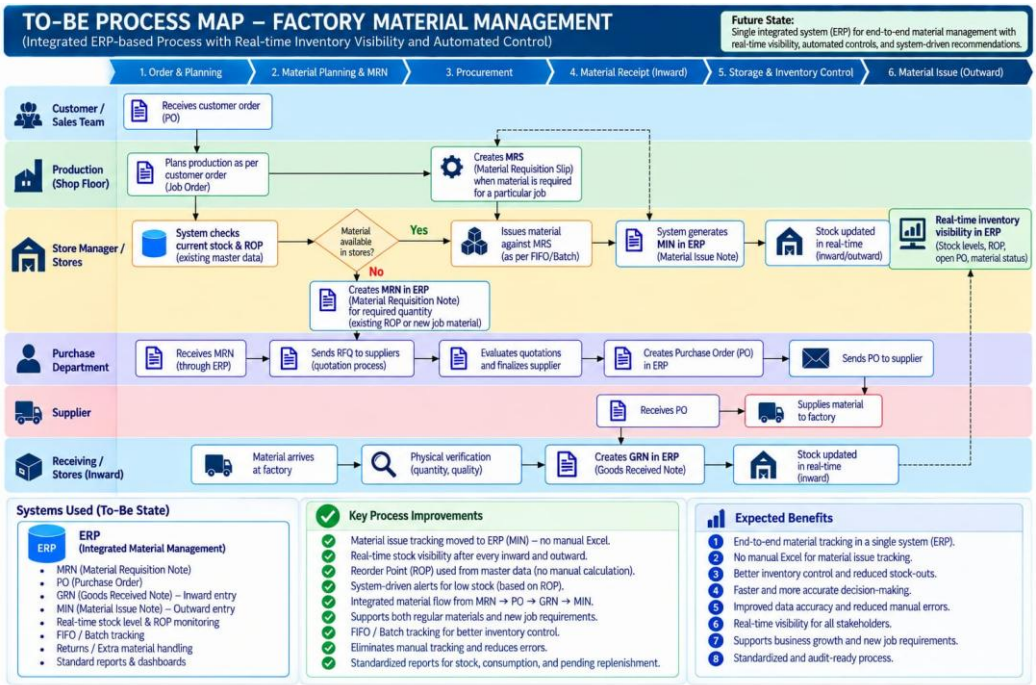


Figure 10 Source reference proposed material management map

Source S3. Original visual retained; source resolution governs legibility.

FSN Class Profiles

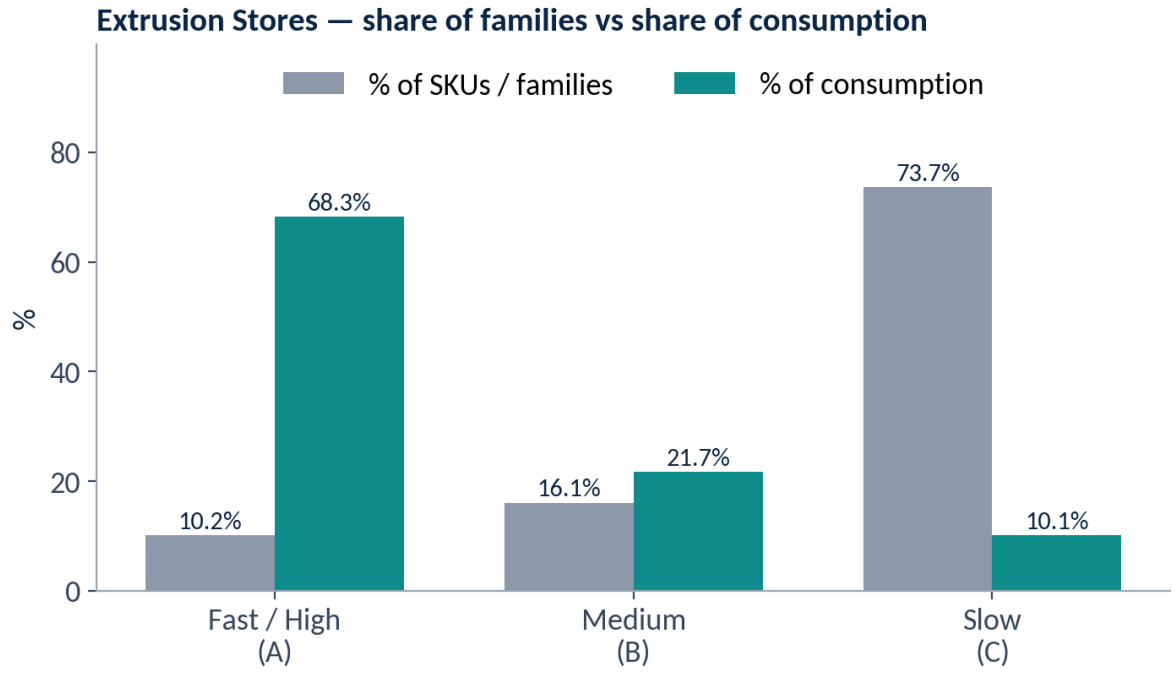


Figure 11 Extrusion families and consumption by class

Source S2. Original visual retained; source resolution governs legibility.

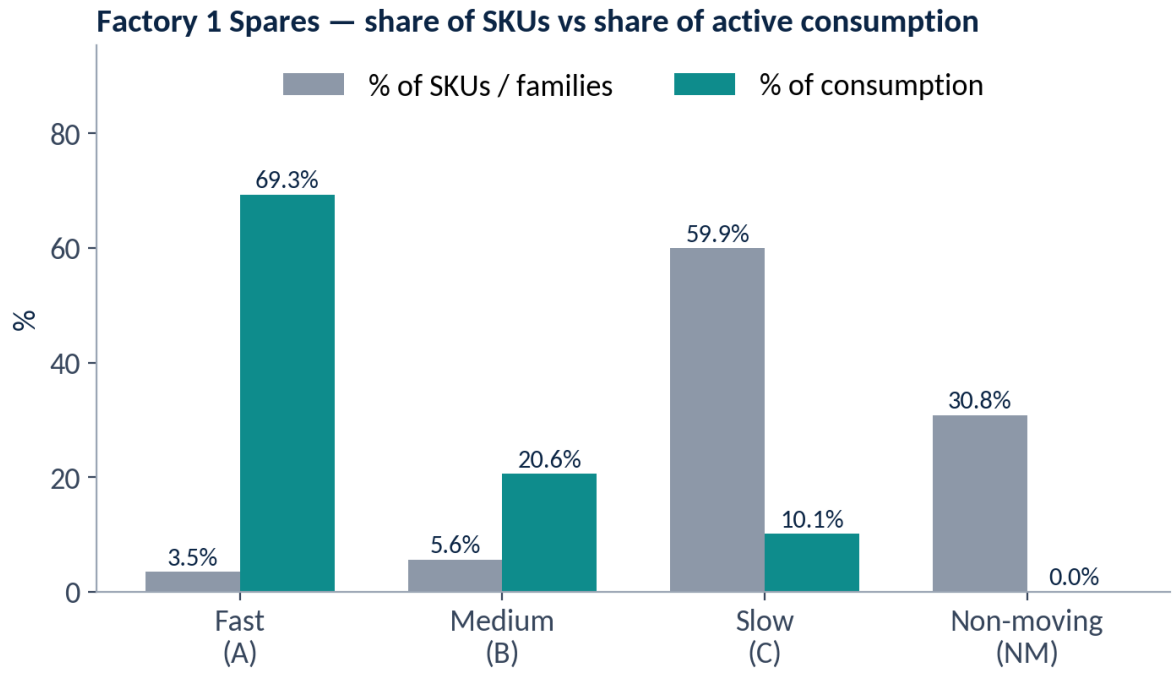


Figure 12 Spares SKUs and active consumption by class

Source S2. Original visual retained; source resolution governs legibility.

FSN Fast Spares and Demand Variation

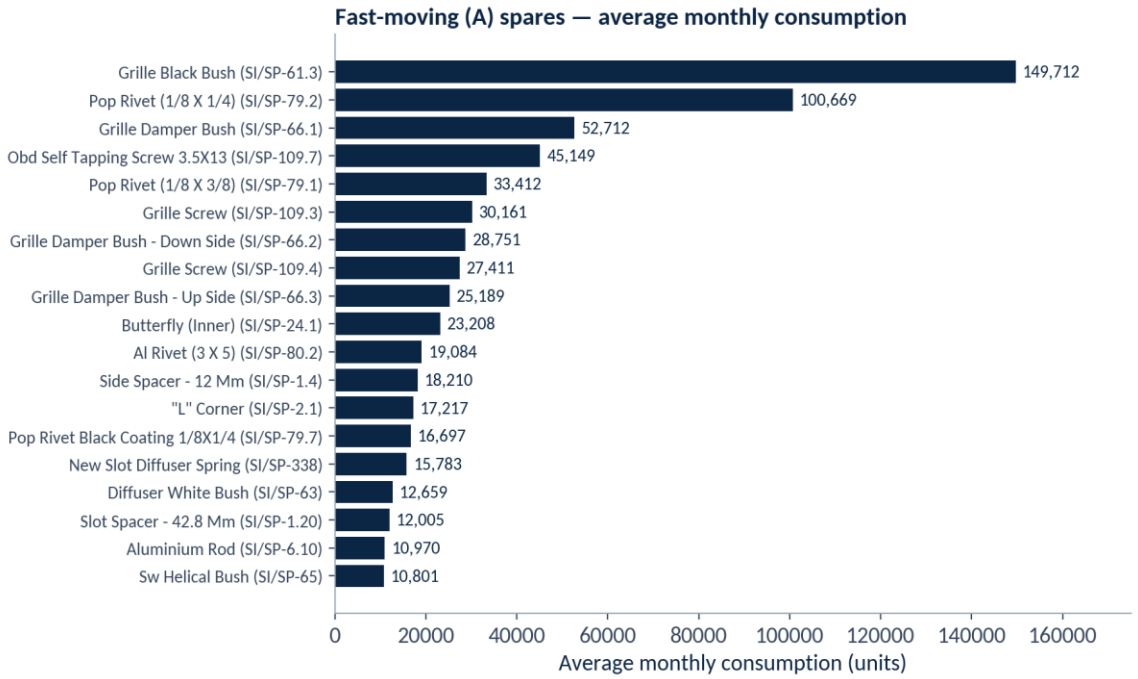


Figure 13 Fast spares average monthly consumption

Source S2. Original visual retained; source resolution governs legibility.

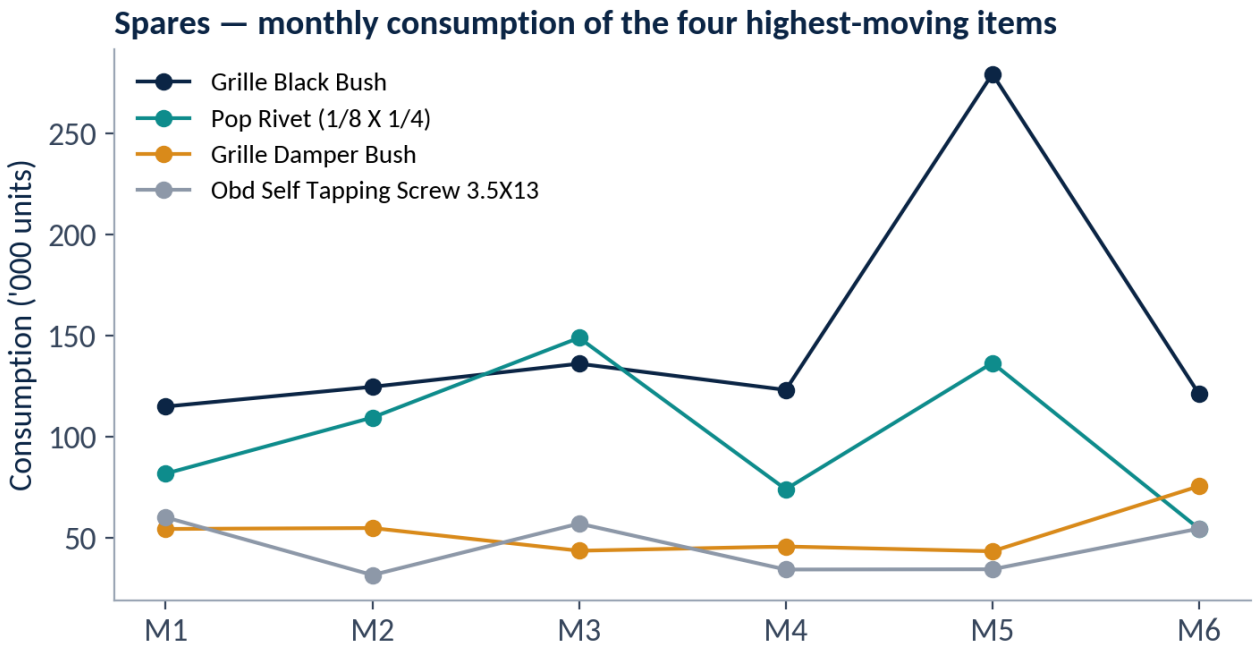


Figure 14 Monthly consumption of four leading spares

Source S2. Original visual retained; source resolution governs legibility.

FSN Variability and Non Moving Spares

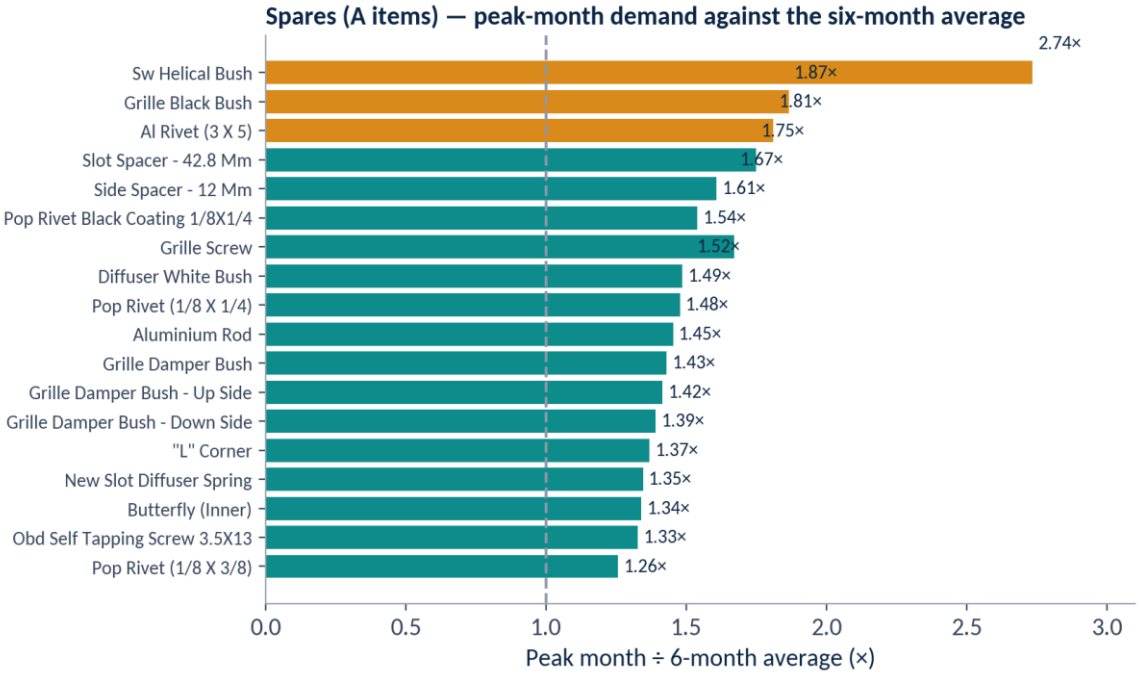


Figure 15 Peak month relative to average for Fast spares

Source S2. Original visual retained; source resolution governs legibility.

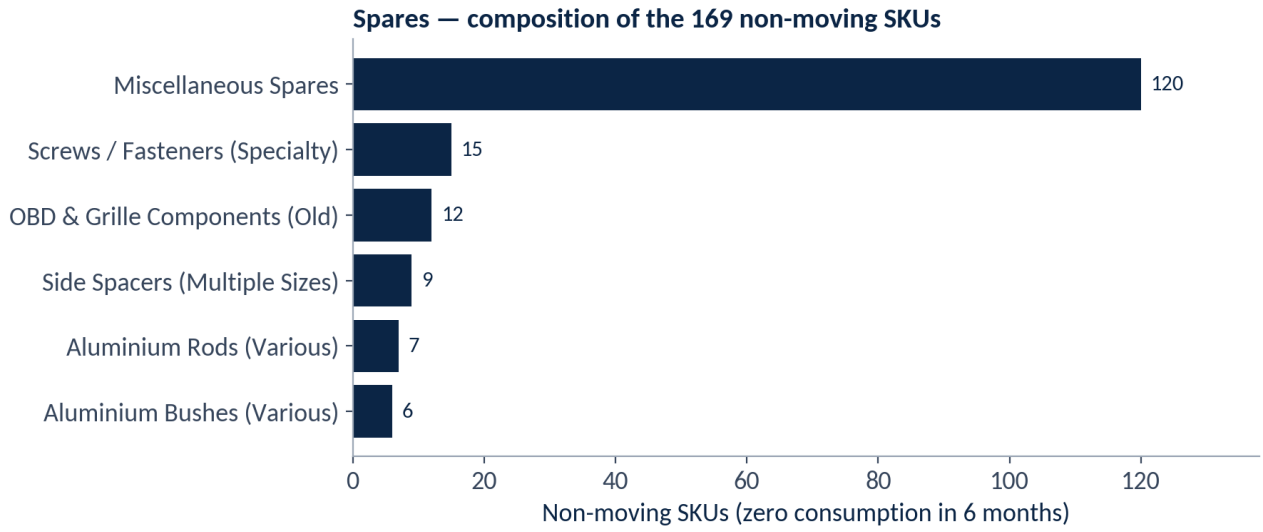


Figure 16 Composition of 169 non moving Spares SKUs

Source S2. Original visual retained; source resolution governs legibility.

FSN Powder Class and Pareto Evidence

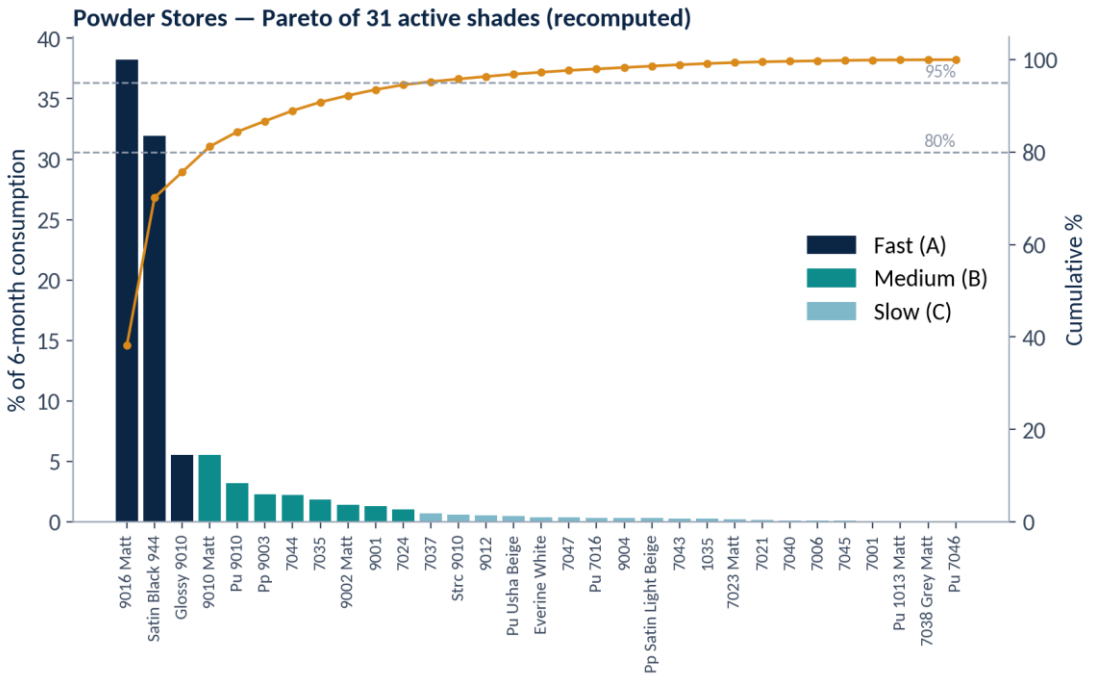


Figure 17 Powder Pareto of 31 shades with consumption

Source S2. Original visual retained; source resolution governs legibility.

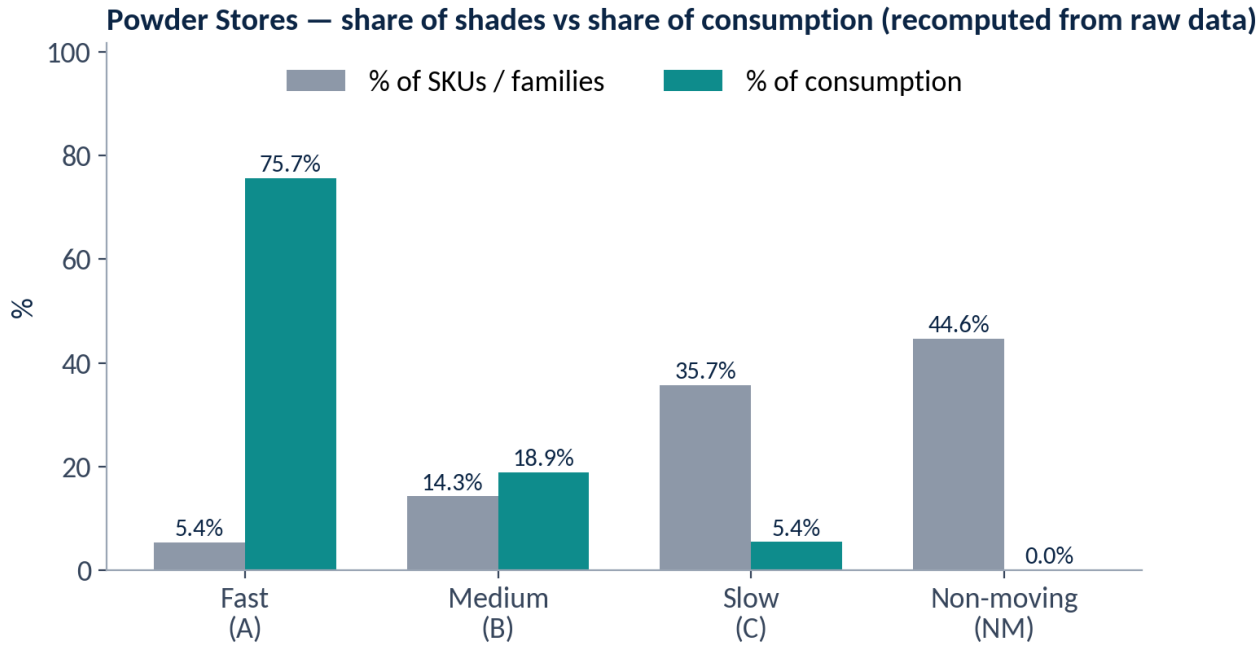


Figure 18 Powder shade and consumption shares by class

Source S2. Original visual retained; source resolution governs legibility.

FSN Powder Pattern and Reorder Structure

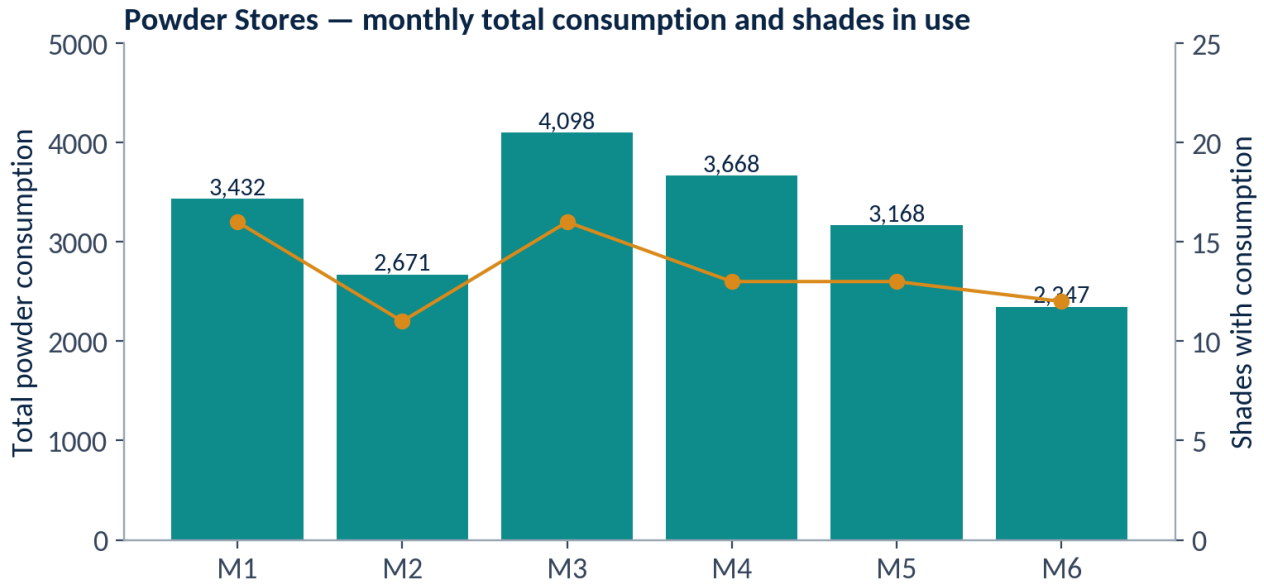


Figure 19 Powder monthly consumption and shades in use

Source S2. Original visual retained; source resolution governs legibility.

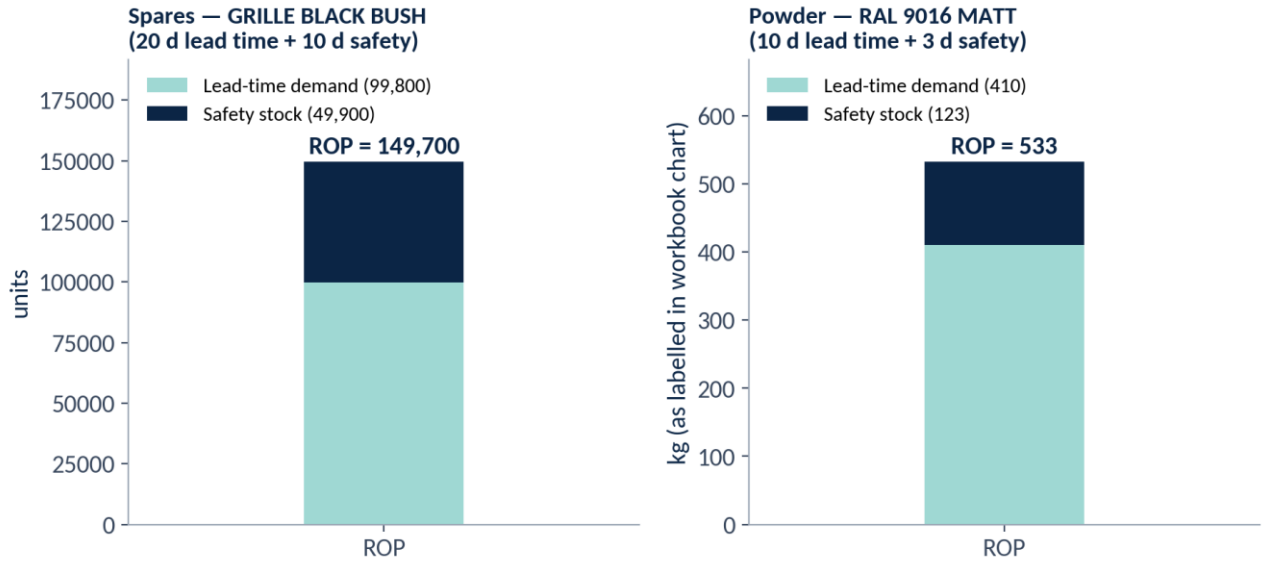


Figure 20 Reorder point worked examples in the FSN source

Source S2. Original visual retained; source resolution governs legibility.

Bin Procurement Source Records

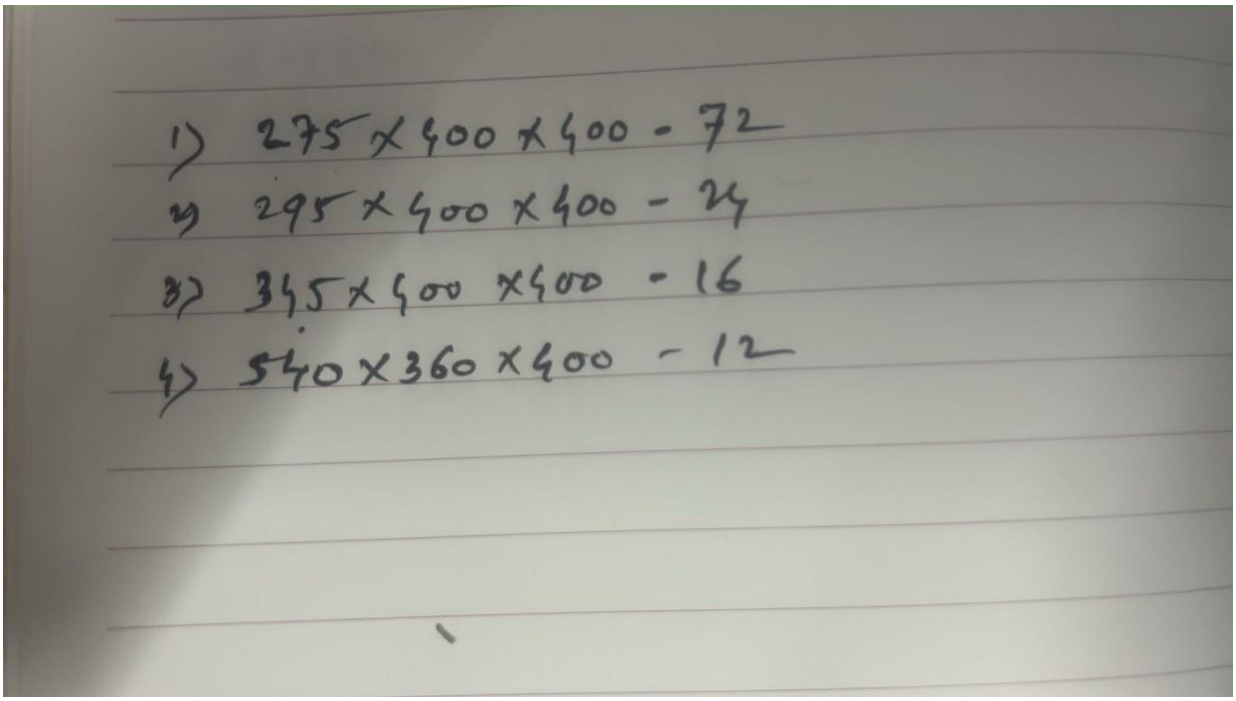


Figure 21 Handwritten bin dimensions and order quantities sheet one

Source S8. Original visual retained; source resolution governs legibility.

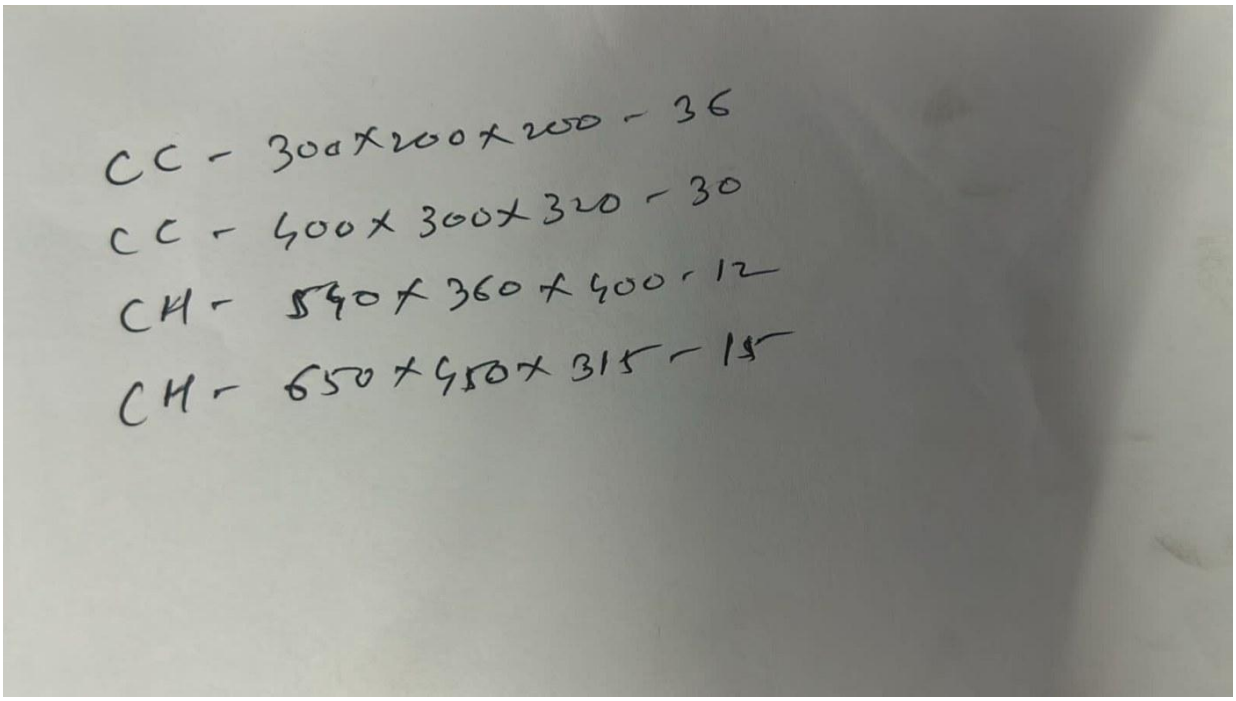


Figure 22 Handwritten bin dimensions and order quantities sheet two

Source S8. Original visual retained; source resolution governs legibility.

Department Rack Arrangement



Figure 23 Blue bins arranged within existing rack

Source S8. Original visual retained; source resolution governs legibility.



Figure 24 Rack and blue bin arrangement second view

Source S8. Original visual retained; source resolution governs legibility.

Location Boards and Material Labels



Figure 25 Rack 5 level boards

Source S8. Original visual retained; source resolution governs legibility.



Figure 26 Round Diffuser Rod bin label and displayed quantity

Source S8. Original visual retained; source resolution governs legibility.

Linear Grille Material Identification



Figure 27 Linear Grille Frame material with label

Source S8. Original visual retained; source resolution governs legibility.



Figure 28 Linear Grille Frame label close view

Source S8. Original visual retained; source resolution governs legibility.

Physical FIFO Bin Arrangement



Figure 29 Spare bin with GI sheet divider

Source S6. Original visual retained; source resolution governs legibility.



Figure 30 Front and rear stock compartments viewed from above

Source S6. Original visual retained; source resolution governs legibility.

Physical FIFO Staff Briefing



Figure 31 Red and green stock identification briefing

Source S6. Original visual retained; source resolution governs legibility.

Other FIFO Material Storage Photographs



Figure 32 Carton storage area with FIFO category to confirm

Source S6. Original visual retained; source resolution governs legibility.



Figure 33 Roll and carton storage area with configuration to confirm

Source S6. Original visual retained; source resolution governs legibility.



Figure 34 Shelved boxed material storage with configuration to confirm

Source S6. Original visual retained; source resolution governs legibility.

Store Kanban Board



Figure 35 Store Kanban Control Board reference implementation

Source S5. Original visual retained; source resolution governs legibility.

The source photograph shows empty pockets. It does not establish card movement or a stockout observation period.

Kanban Card Examples

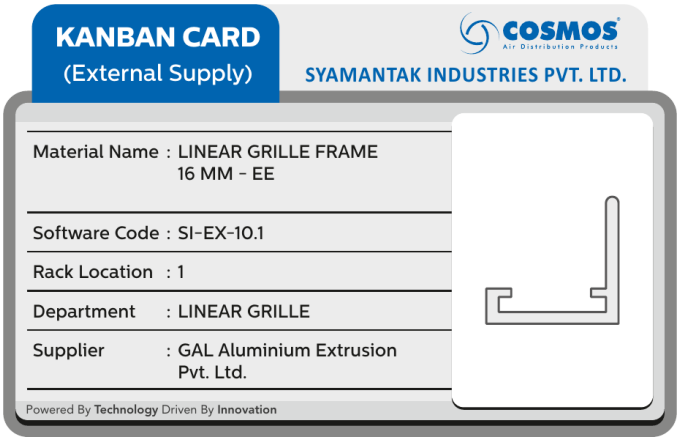


Figure 36 External Supply card for Linear Grille Frame

Source S5. Original visual retained; source resolution governs legibility.

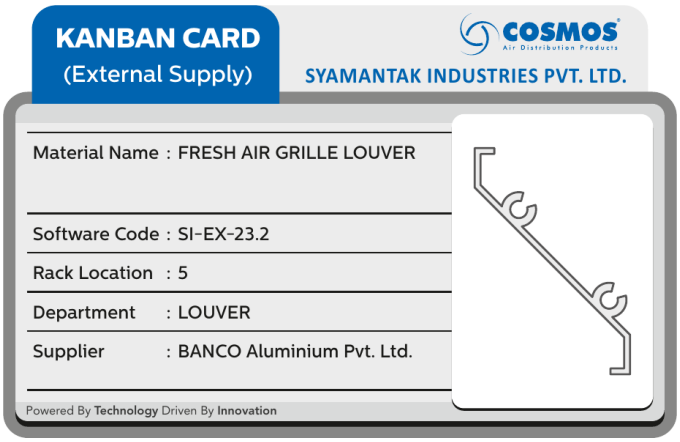


Figure 37 External Supply card for Fresh Air Grille Louver

Source S5. Original visual retained; source resolution governs legibility.

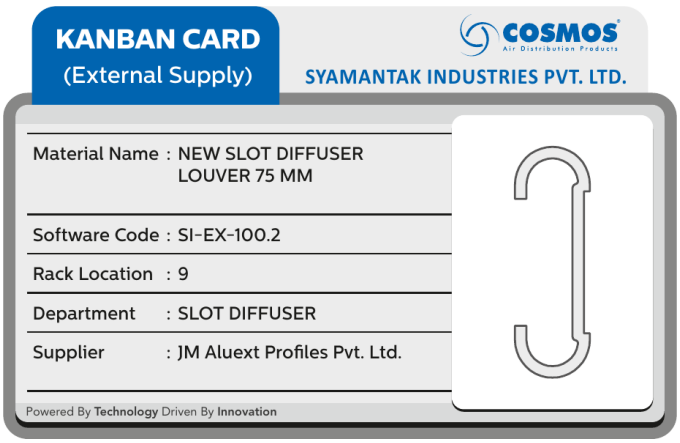


Figure 38 External Supply card for New Slot Diffuser Louver

Source S5. Original visual retained; source resolution governs legibility.

Counting Stopwatch Evidence



Figure 39 Before stopwatch reading 14 minutes 58 seconds

Source S4. Original visual retained; source resolution governs legibility.



Figure 40 After stopwatch reading 7 minutes 53 seconds

Source S4. Original visual retained; source resolution governs legibility.

Counting MRS Evidence

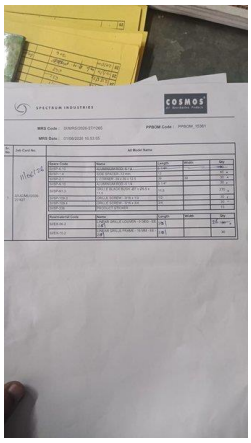


Figure 41 Counting study MRS slip one

Source S4. Original visual retained; source resolution governs legibility.

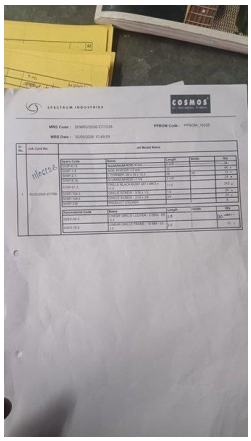


Figure 42 Counting study MRS slip two

Source S4. Original visual retained; source resolution governs legibility.

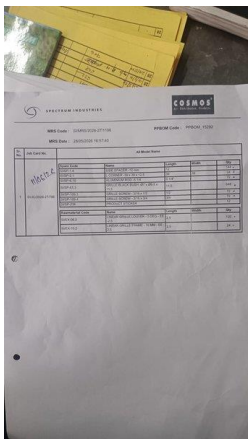


Figure 43 Counting study MRS slip three

Source S4. Original visual retained; source resolution governs legibility.

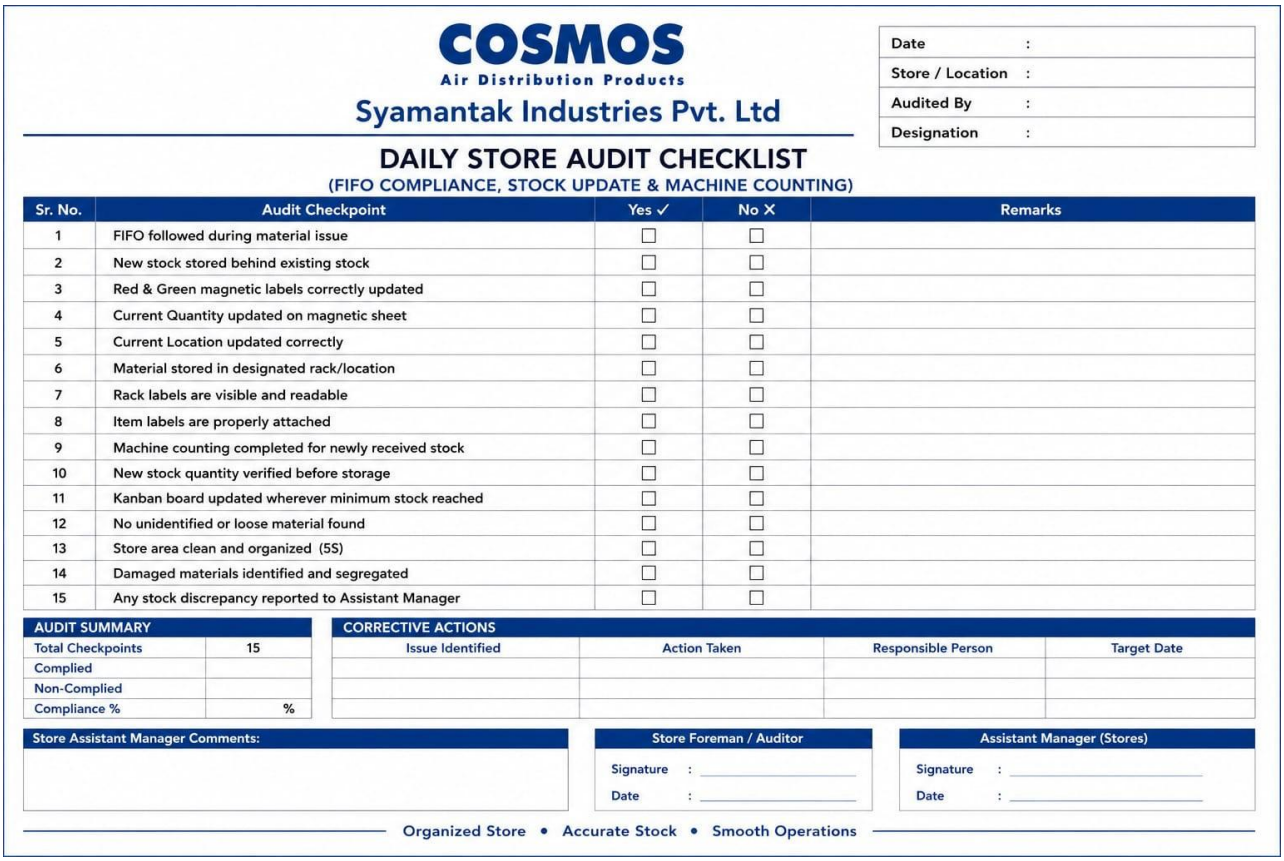
Counting Equipment



Figure 44 Existing digital weighing machine

Source S4. Original visual retained; source resolution governs legibility.

Daily Store Audit Checklist Original



COSMOS
Air Distribution Products
Syamantak Industries Pvt. Ltd

Date	:
Store / Location	:
Audited By	:
Designation	:

DAILY STORE AUDIT CHECKLIST
(FIFO COMPLIANCE, STOCK UPDATE & MACHINE COUNTING)

Sr. No.	Audit Checkpoint	Yes ✓	No ✗	Remarks
1	FIFO followed during material issue	☐	☐	
2	New stock stored behind existing stock	☐	☐	
3	Red & Green magnetic labels correctly updated	☐	☐	
4	Current Quantity updated on magnetic sheet	☐	☐	
5	Current Location updated correctly	☐	☐	
6	Material stored in designated rack/location	☐	☐	
7	Rack labels are visible and readable	☐	☐	
8	Item labels are properly attached	☐	☐	
9	Machine counting completed for newly received stock	☐	☐	
10	New stock quantity verified before storage	☐	☐	
11	Kanban board updated wherever minimum stock reached	☐	☐	
12	No unidentified or loose material found	☐	☐	
13	Store area clean and organized (5S)	☐	☐	
14	Damaged materials identified and segregated	☐	☐	
15	Any stock discrepancy reported to Assistant Manager	☐	☐	

AUDIT SUMMARY	
Total Checkpoints	15
Complied	
Non-Complied	
Compliance %	%

CORRECTIVE ACTIONS			
Issue Identified	Action Taken	Responsible Person	Target Date

Store Assistant Manager Comments:	Store Foreman / Auditor	Assistant Manager (Stores)
	Signature : _____	Signature : _____
	Date : _____	Date : _____

Organized Store • Accurate Stock • Smooth Operations

Figure 45 Original Daily Store Audit Checklist

Source S4. Original visual retained; source resolution governs legibility.

FIFO Visual Work Instruction English

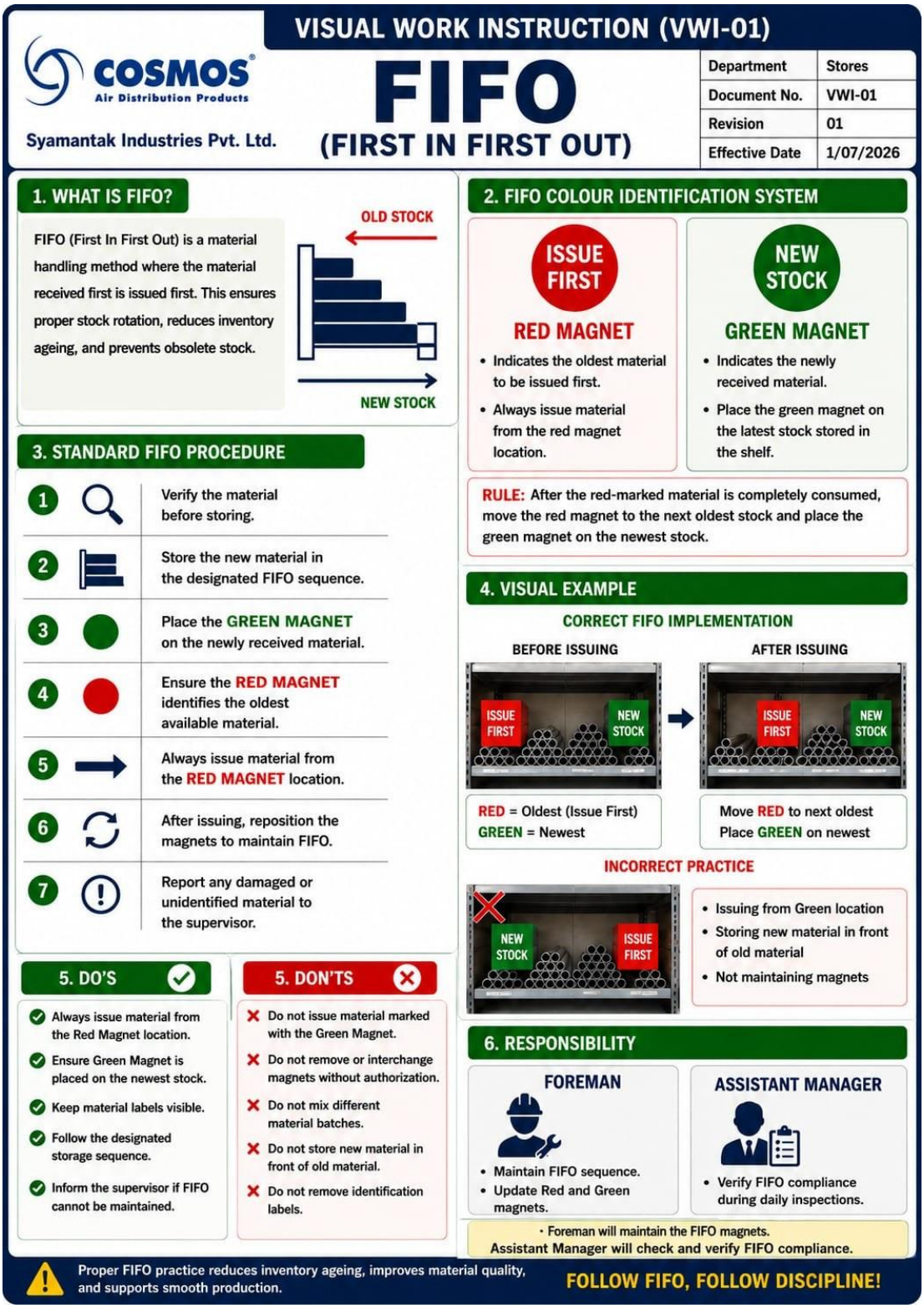


Figure 46 VWI 01 FIFO English Revision 01 effective 01 July 2026

Source S4. Original visual retained; source resolution governs legibility.

FIFO Visual Work Instruction Marathi

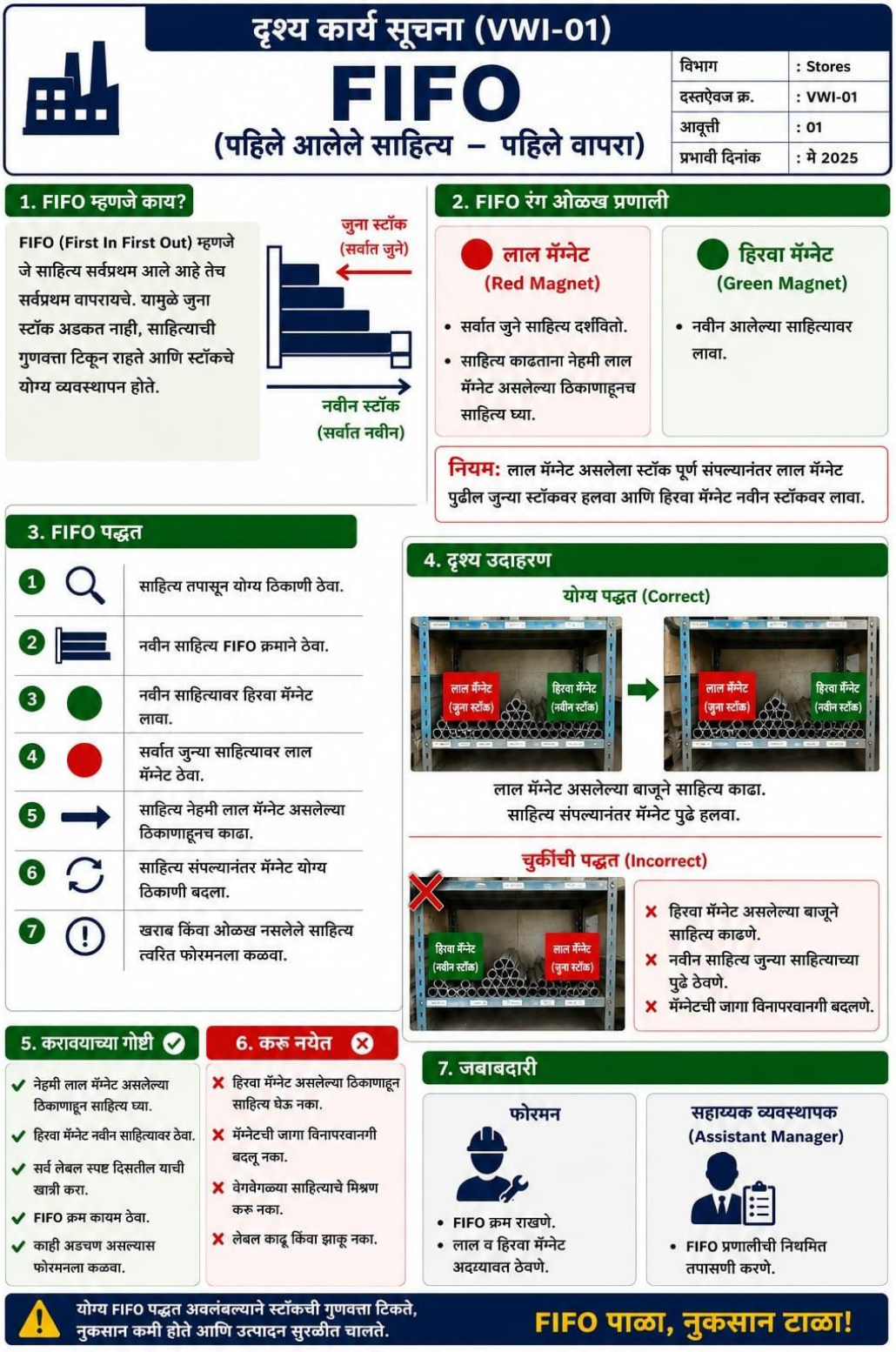


Figure 47 VWI 01 FIFO Marathi Revision 01 effective 01 July 2026

Source S4. Original visual retained; source resolution governs legibility.

Kanban Visual Work Instruction English

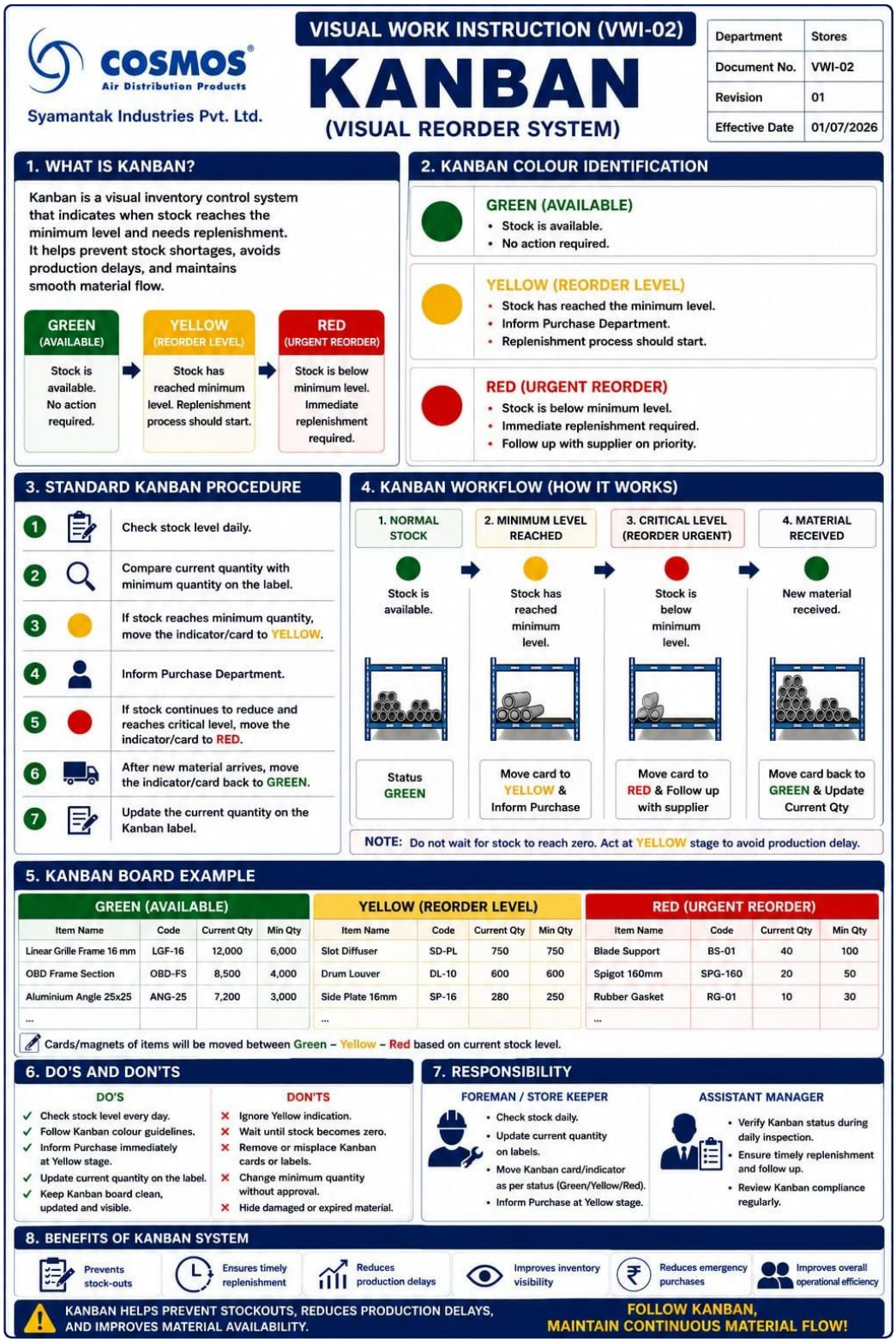


Figure 48 VWI 02 Kanban English Revision 01 effective 01 July 2026

Source S4. Original visual retained; source resolution governs legibility.

Kanban Visual Work Instruction Marathi

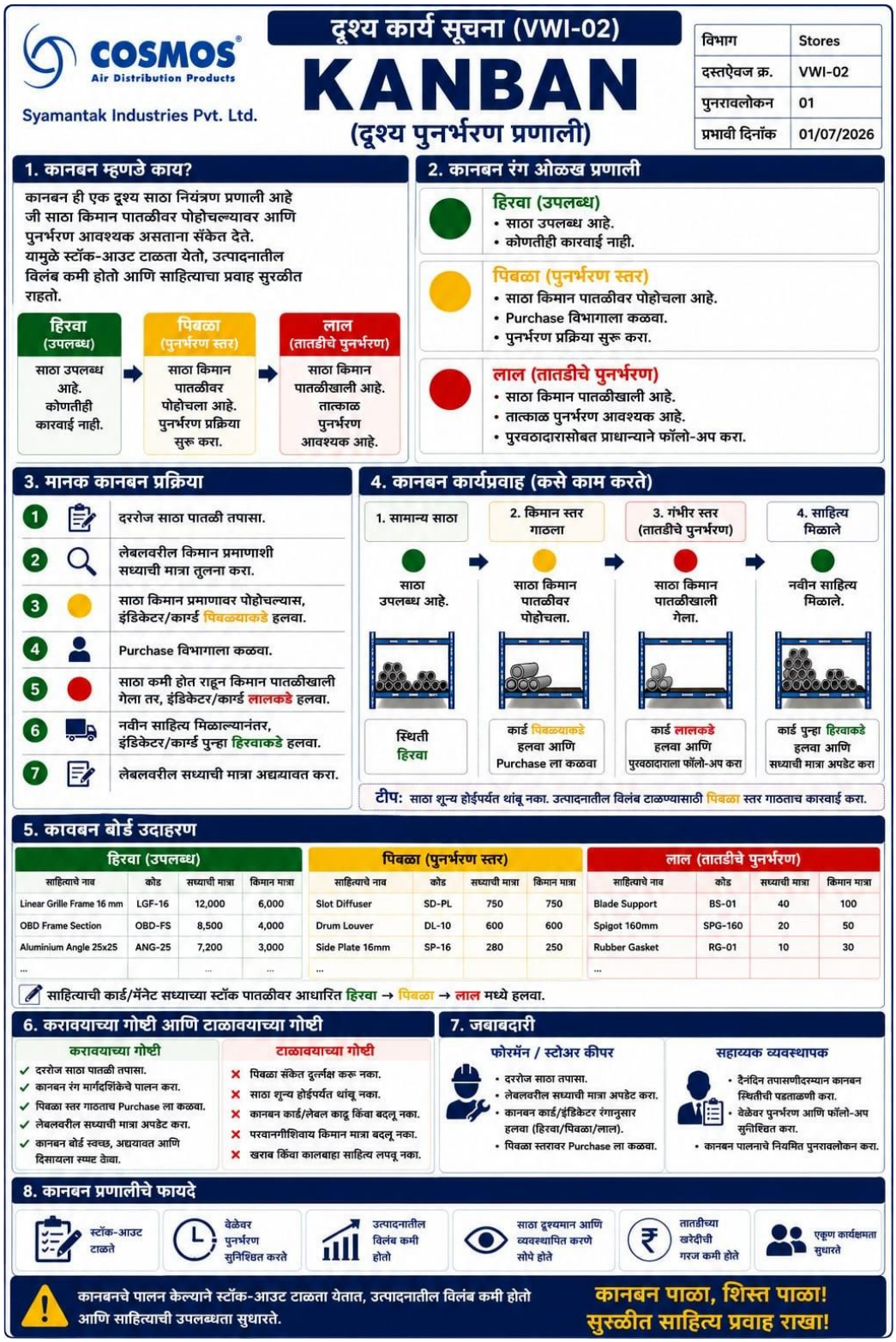


Figure 49 VWI 02 Kanban Marathi Revision 01 effective 01 July 2026

Source S4. Original visual retained; source resolution governs legibility.

VAV Original Layout Photograph

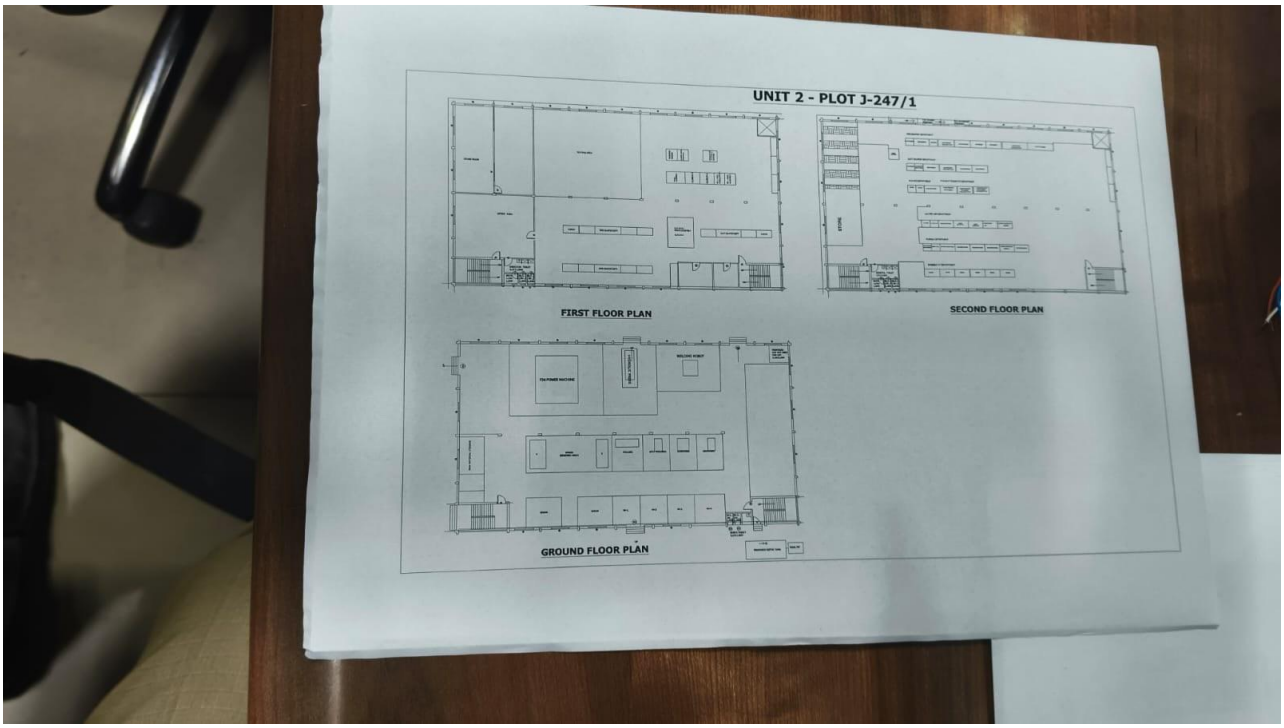


Figure 50 Original multi floor layout sheet Unit 2 Plot J 247 1

Source S4. Original visual retained; source resolution governs legibility.

The drawing is the primary visual record. Retained source resolution limits some labels; no geometry or flow arrow has been invented.

VAV First Floor Layout

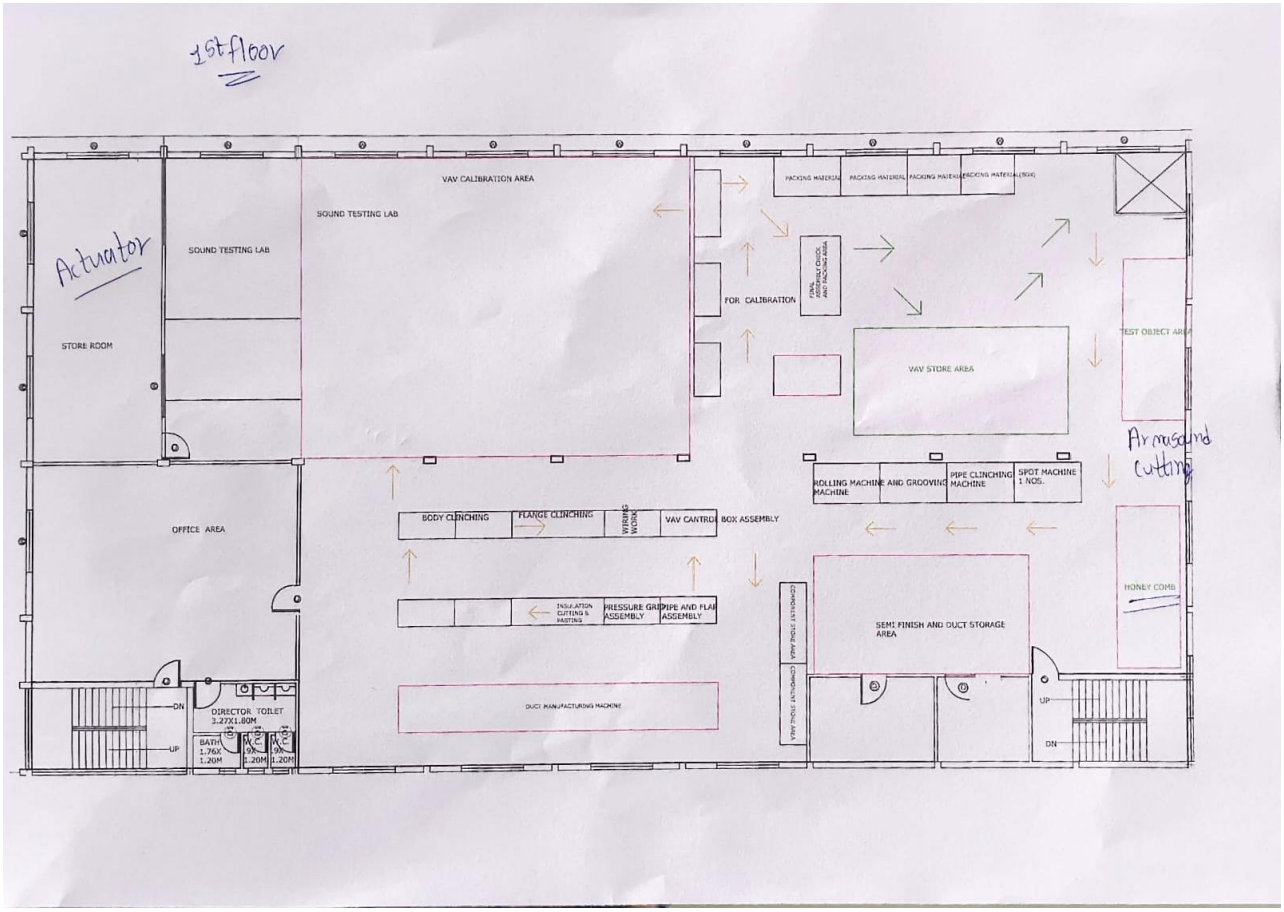


Figure 51 First floor layout with VAV and hand drawn flow arrows

Source S4. Original visual retained; source resolution governs legibility.

The drawing is the primary visual record. Retained source resolution limits some labels; no geometry or flow arrow has been invented.

VAV Ground Floor Plan

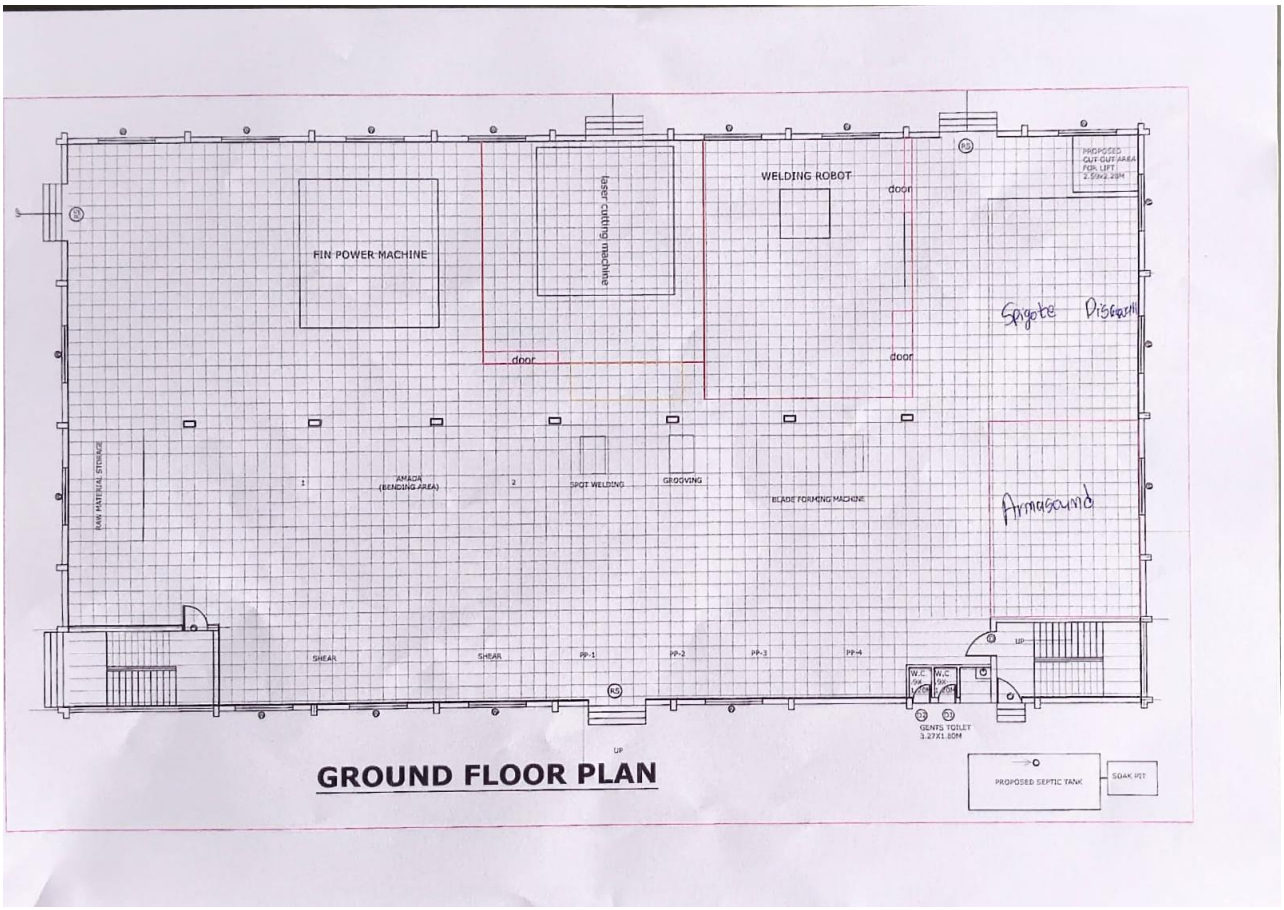


Figure 52 Ground floor plan retained from source

Source S4. Original visual retained; source resolution governs legibility.

The drawing is the primary visual record. Retained source resolution limits some labels; no geometry or flow arrow has been invented.

VAV Second Floor Layout

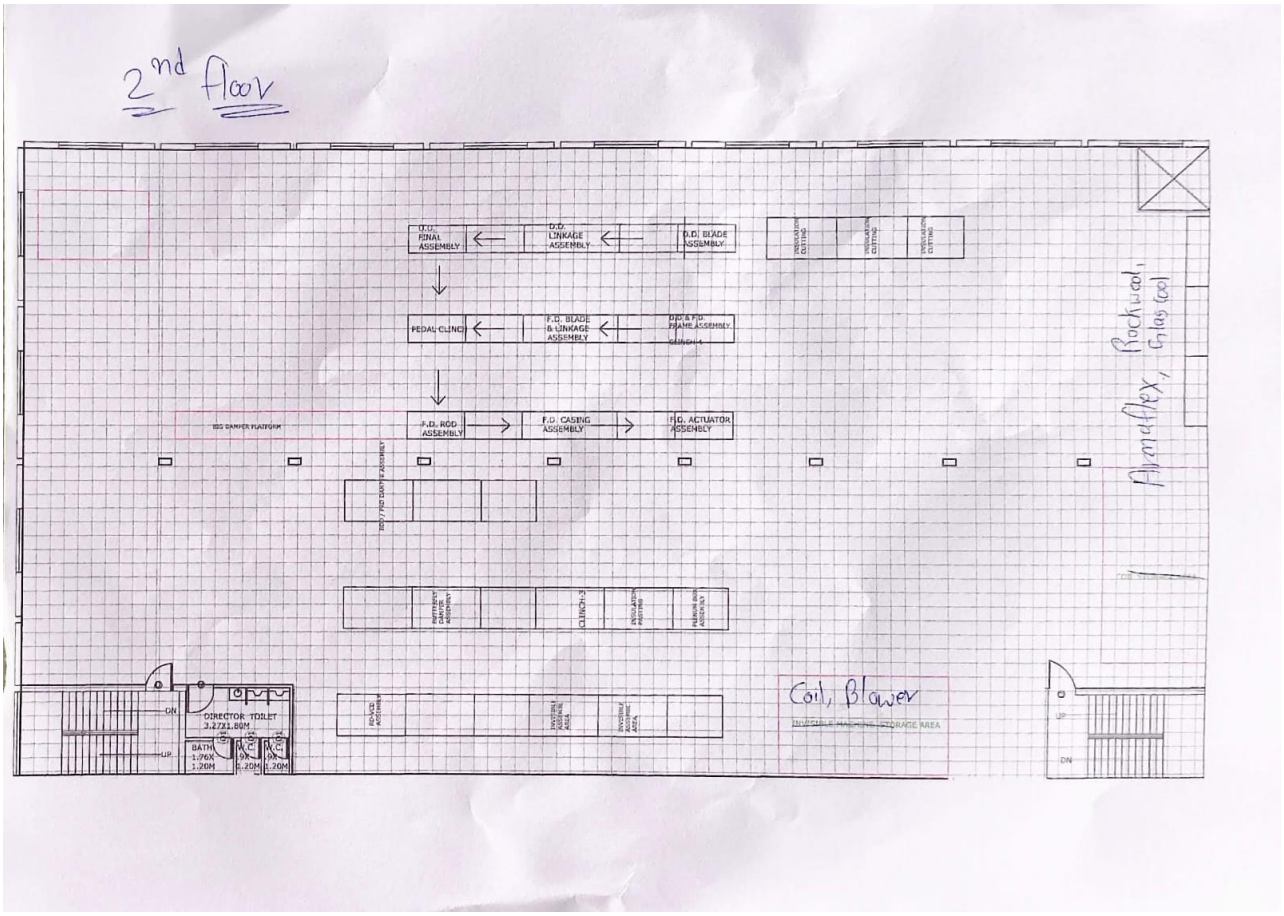


Figure 53 Second floor layout retained from source

Source S4. Original visual retained; source resolution governs legibility.

The drawing is the primary visual record. Retained source resolution limits some labels; no geometry or flow arrow has been invented.

Bin Box Visualisation Photographs



Figure 54 Rack level material boards overview

Source S4. Original visual retained; source resolution governs legibility.



Figure 55 Labelled bins in the walkthrough still

Source S4. Original visual retained; source resolution governs legibility.

Bin Box Earlier Condition



Figure 56 Earlier rack condition before material boards

Source S4. Original visual retained; source resolution governs legibility.

Mezzanine Flooring Evidence



Figure 57 Wooden flooring with local damage

Source S4. Original visual retained; source resolution governs legibility.

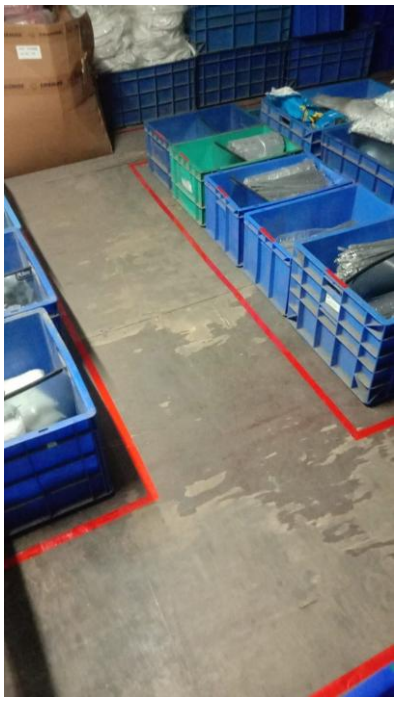


Figure 58 Current mezzanine floor and red tape marking

Source S4. Original visual retained; source resolution governs legibility.

Current-floor imagery does not certify material specification, load capacity or engineering suitability.

Appendix B Source Process Drawings and Walkthrough Objects

The original kitting process drawings and source embedded walkthrough objects follow. Their status is governed by the narrative: kitting remains proposed. Walkthrough playback depends on Word support and local security settings; static photographs elsewhere provide the visible evidence.

Original Kitting Process Drawing 1

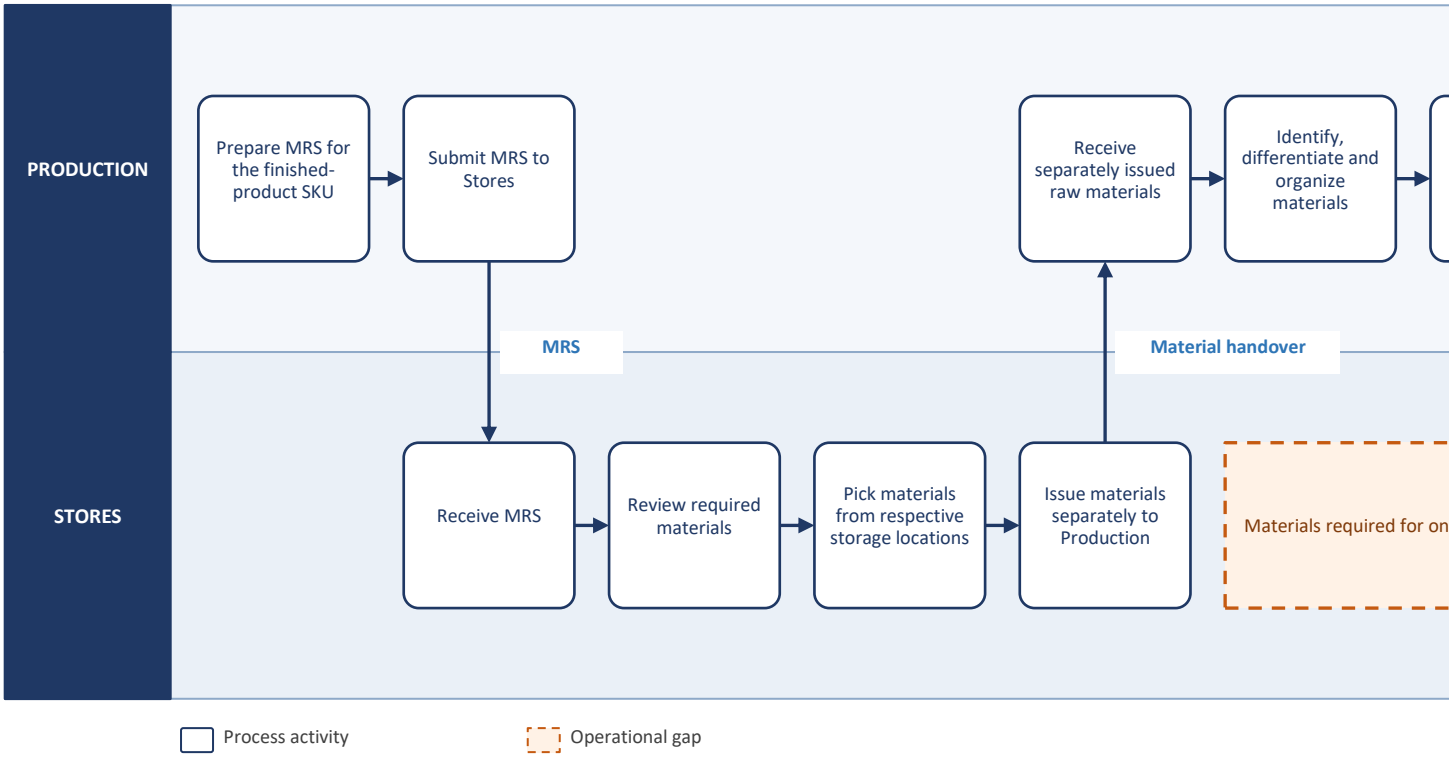


Figure 59 Original kitting process drawing retained from S7

Original Kitting Process Drawing 2

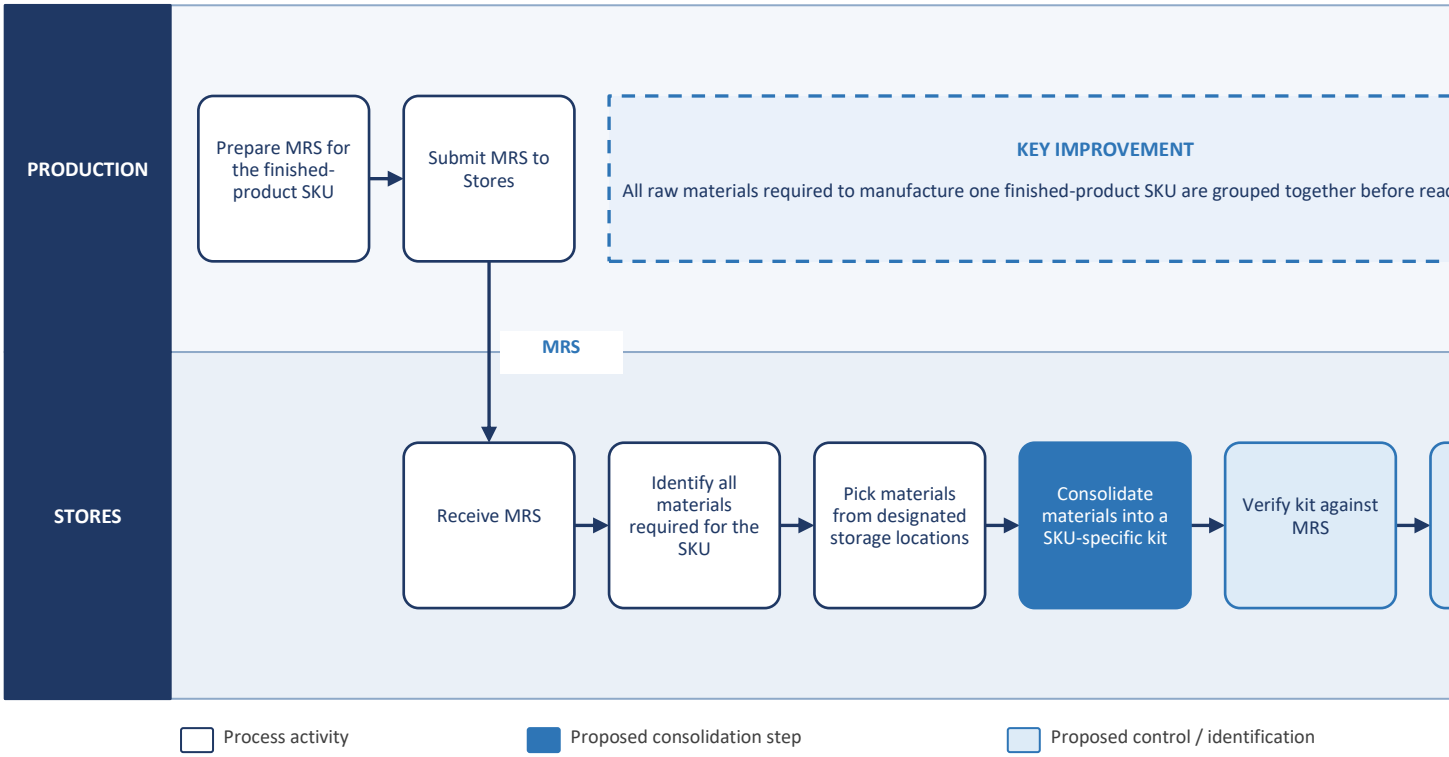


Figure 60 Original kitting process drawing retained from S7

Retained Walkthrough Object from S6



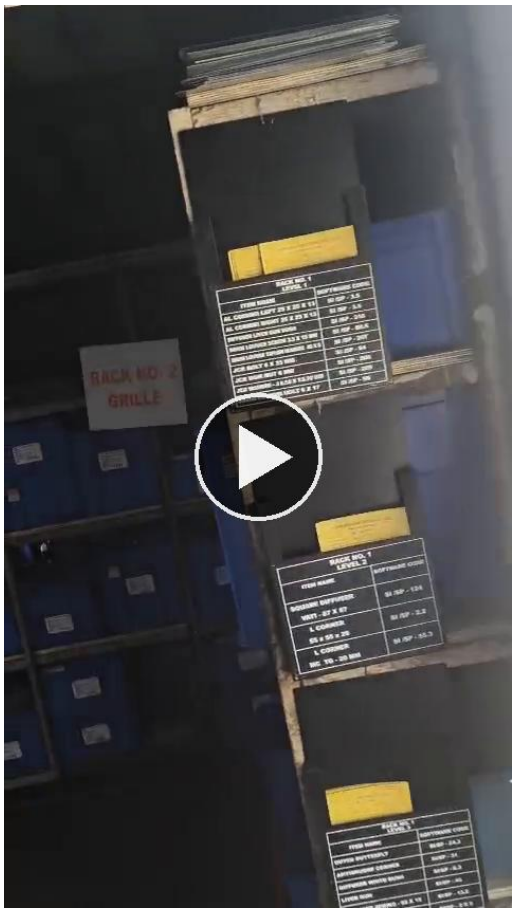
Original embedded object and poster retained from S6. Playback is application dependent.

Retained Walkthrough Object from S6



Original embedded object and poster retained from S6. Playback is application dependent.

Retained Walkthrough Object from S4



Original embedded object and poster retained from S4. Playback is application dependent.